Mathematical and Physical Modeling of Entangled Photon Propagation in Optical Fibers

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May, 2026

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Chapter 1

General Framework

1.1 Purpose and Scope

The purpose of this work is to establish a coherent framework for the mathematical modeling and numerical simulation of polarization-entangled photon pairs propagating through optical fibers.
The central objective is to organize the physical assumptions, mathematical formalism, and simulation strategy into a reproducible structure, so that each computational step can be traced back to a precise theoretical element. In this way, the simulator is not developed as an isolated numerical tool, but as a direct operational translation of the underlying quantum model.

The modeling approach adopted throughout this work is intentionally progressive. Simplified analytically tractable models are introduced first, and are then extended toward increasingly realistic descriptions incorporating statistical effects, decoherence, and effective noisy evolution.
This strategy enables a controlled connection between physical interpretation, analytical predictions, and numerical validation.

1.2 Physical Scenario and Experimental Setting

The physical system considered in this work consists of a source of polarization-entangled photon pairs, two independent optical-fiber arms, and a final polarization-measurement stage.
A spontaneous parametric down-conversion (SPDC) source generates photon pairs ideally prepared in a Bell state. The polarization degree of freedom encodes the qubit associated with each photon, with the horizontal and vertical polarization modes identified as:

$$\{|H\rangle, |V\rangle\}$$

After generation, the two photons are separated into two spatially distinct paths, denoted A and B, and propagate through independent optical fibers.

The fibers affect the polarization state of the propagating photons. In the experimental setting considered here, the relevant propagation effects are represented by two main physical mechanisms.

The first one is a residual coherent phase shift between the horizontal and vertical polarization components. After partial polarization compensation, the remaining deterministic or slowly varying effect is modeled as a relative phase accumulated during propagation. Physically, this phase originates from birefringence, optical-path differences, mechanical perturbations, and thermal fluctuations along the fiber. At the two-photon level, this modifies the phase relation between the components of the entangled state and therefore changes the measured polarization correlations.

The second mechanism is local depolarization. While the phase shift describes a coherent transformation of the polarization state, depolarization represents an effective loss of polarization information in each fiber arm. This accounts for unresolved coupling to additional degrees of freedom, imperfect compensation, environmental instability, or fluctuations that are not explicitly tracked at the measurement stage. Since the two photons propagate through different fibers, this effect is treated locally and independently on arms A and B.

At the output of the fibers, the photons are collected by polarization analyzers and single-photon detectors. Measurements are performed on the two arms jointly, so that the relevant experimental information is contained in coincidence statistics rather than in single-arm detection probabilities alone.
From these joint measurements one extracts polarization correlations and state-level metrics.

The physical scenario can therefore be summarized as:

$$\begin{gathered}\text{SPDC entangled-photon source}\\ \downarrow\\ \text{Separation into fiber arms } A \text{ and } B\\ \downarrow\\ \text{Residual phase shift and local depolarization}\\ \downarrow\\ \text{Polarization analysis and coincidence detection}\\ \downarrow\\ \text{Extraction of correlations and entanglement metrics.}\end{gathered}$$

1.3 Theoretical Formalism and Notation

The physical scenario described above is formulated within the language of finite-dimensional quantum information theory. The two photons are treated as a bipartite quantum system, where each subsystem carries one polarization qubit and the joint state belongs to the tensor-product space associated with arms A and B.

The notation follows, as far as possible, the standard conventions of quantum information theory, in particular those used in *Quantum Computation and Quantum Information* by Nielsen and Chuang [1]. Pure states are denoted by kets such as $|\psi\rangle$, density operators by ρ, local subsystems by labels A and B, and tensor products by $\otimes$.

Single-photon polarization states are expressed in the computational basis identified with the horizontal and vertical polarization modes,

$$|H\rangle \equiv |0\rangle, \qquad |V\rangle \equiv |1\rangle.$$

The corresponding two-photon basis is written as

$$\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\},$$

where the first label refers to arm A and the second label refers to arm B.

Coherent polarization transformations are described by local operators acting on the individual arms, while effective noisy processes are described at the density-operator level. This allows the same formalism to treat pure-state evolution, statistical mixtures, and non-unitary propagation effects.

Observable quantities are expressed through expectation values of Hermitian operators. In particular, polarization correlations are described using Pauli operators acting locally on the two arms.

The propagated state is characterized through a fixed set of quantities used throughout the work: fidelity F, purity γ, concurrence C, tangle τ, entanglement of formation E_F, and polarization correlations $\langle \sigma_i \otimes \sigma_j \rangle$. Their formal definitions are introduced in the dedicated theoretical chapters, while Chapter 1 only establishes the notation used to refer to them.

1.4 Computational Architecture and Framework

The computational framework developed in this work is designed as a modular numerical counterpart of the theoretical description. Its purpose is to make each physical or mathematical object appearing in the text correspond to a clear computational component in the codebase.

The simulator is organized into functional layers for state preparation, operator construction, state evolution, quantum channels, measurement evaluation, metric computation, testing, and experiment-specific workflows. This organization separates reusable low-level routines from higher-level numerical experiments, improving reproducibility and making each result traceable to the theoretical object from which it originates.

State routines construct Bell states and density matrices; evolution routines apply local unitary transformations; channel routines implement global, local, two-arm, and composite fiber-channel maps; measurement routines evaluate correlation functions, correlation tensors, and CHSH quantities; metric routines compute the scalar indicators used to characterize the propagated state.

Numerical experiments are built as higher-level workflows that combine these validated components. Each experiment specifies an input state, a propagation mechanism, a set of observables or metrics, and a comparison with analytical predictions whenever available. This makes the simulator not only a numerical tool, but also a reproducibility layer connecting physical assumptions, mathematical formulation, and numerical results.

The role of the computational architecture is therefore to support a one-to-one correspondence between theoretical modeling and numerical implementation. Each output of the simulator should be interpretable in terms of a precise state preparation, propagation mechanism, measurement operator, or metric definition.

Chapter 2

Modeling Strategy

2.1 Progressive Modeling Philosophy

The modeling approach adopted in this work follows a progressive refinement strategy. The physical system is not introduced directly through the most complete effective description. Instead, the fiber model is constructed by adding one propagation mechanism at a time, so that the role of each contribution can be isolated analytically and validated numerically.

The central physical object is an entangled photon pair distributed through two independent optical-fiber arms. The fiber action is described at the level of the polarization qubits. Within this setting, the relevant effects are organized into coherent phase evolution, statistical phase fluctuations, and depolarizing noise.

This hierarchy serves two purposes. From the theoretical point of view, it separates distinct physical mechanisms and avoids mixing coherent, statistical, and noisy effects at the same modeling level. From the computational point of view, it leads to a modular simulator architecture, where each new model extends a previously validated implementation.

The first levels of the hierarchy provide analytically tractable reference cases. Later levels introduce effective noise mechanisms and composite fiber-channel descriptions, progressively improving the representation of two photons propagating through independent optical fibers.

2.2 Modeling Hierarchy

The starting point is the ideal Bell-state preparation. This defines the reference entangled state against which all propagation and degradation effects are evaluated.

The model is then refined through successive levels:

$$\begin{array}{c}\text{Ideal Bell-state preparation}\\ \downarrow \\ \text{Fixed reduced phase evolution}\\ \downarrow \\ \text{Random phase ensemble}\\ \downarrow \\ \text{Global depolarizing baseline}\\ \downarrow \\ \text{Local two-arm depolarizing channel}\\ \downarrow \\ \text{Composite effective fiber channel}\end{array}$$

The first propagation model is the fixed reduced phase model. After polarization compensation, the dominant residual coherent effect is represented as a relative phase shift between the Bell-state components. This model preserves the purity of the global state and isolates the phase dependence of the polarization correlations.

The next refinement introduces statistical phase fluctuations. Instead of assigning a single fixed value to the phase parameter, the state is averaged over a distribution of sampled phase realizations. This leads naturally to a density-operator description and models loss of coherence due to unresolved phase variability.

Depolarizing noise is then introduced as an effective description of polarization degradation. The global depolarizing channel is used as an analytically simple isotropic-noise baseline, while the local two-arm depolarizing channel acts independently on arms A and B, reflecting the physical geometry of two photons propagating through distinct fibers.

The final developed model is the composite effective fiber channel. It combines residual phase evolution, local two-arm depolarization, and, when required, statistical phase averaging. This model represents the main effective fiber-propagation description considered in the numerical experiments and leaves the framework open to future extensions.

2.3 From Model Levels to Numerical Experiments

Each model level is associated with a dedicated numerical experiment. The experiments are not independent demonstrations, but successive validation stages of the same modeling framework.

This organization makes the numerical part of the thesis cumulative. Each experiment introduces a single additional modeling ingredient with respect to the previous ones, while preserving the same computational structure: state preparation, evolution, observable evaluation, metric computation, and comparison with analytical predictions.

The first experiments establish reference cases in which analytical predictions can be derived explicitly and used to validate the simulator. Later experiments reuse the same validated components in more structured models, where coherent phase evolution, statistical averaging, and effective noise are combined.

Exp.	**Model level**	**New insertion**
1	Reduced phase model	Deterministic phase sweep
2	Random phase ensemble	Statistical averaging over θ_k
3	Global depolarization	Isotropic mixed-state baseline
4	CHSH analysis	Bell-inequality diagnostics from T
5	Composite fiber channel	Phase plus local two-arm depolarization

Table 2.1: Connection between the modeling hierarchy and the numerical experiments. Each experiment introduces only the additional mechanism required at that stage.

This structure ensures that each numerical result can be interpreted as the consequence of a specific modeling refinement. The analytical predictions developed in the theoretical chapters are therefore tested progressively before being combined in the composite fiber-channel model.

Chapter 3

Physical and Mathematical Representation

3.1 Quantum System and Hilbert Space Structure

The physical system considered consists of polarization-entangled photon pairs generated by spontaneous parametric down-conversion (SPDC). The two photons propagate along distinct optical paths, denoted A and B, forming a bipartite quantum system.

In this work, the qubit is encoded in the polarization degree of freedom:

$$|H\rangle \equiv |0\rangle, \qquad |V\rangle \equiv |1\rangle. \tag{3.1}$$

Accordingly, the computational basis of the bipartite system is

$$\mathcal{B}_{AB} = \{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}. \tag{3.2}$$

In polarization notation, this corresponds to

$$\{|H\rangle_A|H\rangle_B, |H\rangle_A|V\rangle_B, |V\rangle_A|H\rangle_B, |V\rangle_A|V\rangle_B\}.$$

The Hilbert space of the total system is therefore

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B \simeq \mathbb{C}^2 \otimes \mathbb{C}^2. \tag{3.3}$$

Since each subsystem is two-dimensional, the composite Hilbert space has dimension 4.
As a consequence:

- pure states are represented by complex vectors of length 4,
- operators acting on the full bipartite system are represented by 4×4 matrices.

This construction provides a concrete realization of the quantum-information framework introduced above.
The fixed computational basis in Eq. (3.2) will be used consistently throughout the theoretical development, the numerical implementation, and the validation experiments.

3.2 Bell-State Initialization

As a reference initial condition, this work considers the Bell state

$$|\Psi^{+}\rangle = \frac{1}{\sqrt{2}} \left(|H\rangle_A |V\rangle_B + |V\rangle_A |H\rangle_B \right). \tag{3.4}$$

Using the computational-basis convention introduced in Eq. (3.1), the same state can be written as

$$|\Psi^{+}\rangle = \frac{1}{\sqrt{2}} \left(|01\rangle + |10\rangle \right). \tag{3.5}$$

In the ordered computational basis $\mathcal{B}_{AB}$, this state is represented by the column vector

$$|\Psi^{+}\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}. \tag{3.6}$$

The state $|\Psi^{+}\rangle$ is maximally entangled and provides the natural input for the baseline version of the simulator. It also serves as the reference state with respect to which later quantities such as fidelity and Bell-inequality violation may be evaluated.

The Bell state provides both a clean analytical benchmark and a physically relevant entangled resource, enabling direct interpretation of fiber-induced effects in terms of coherence, two-photon correlations, and nonlocality.

For completeness, the four Bell states are

$$|\Psi^{\pm}\rangle = \frac{1}{\sqrt{2}} \left(|01\rangle \pm |10\rangle \right), \qquad |\Phi^{\pm}\rangle = \frac{1}{\sqrt{2}} \left(|00\rangle \pm |11\rangle \right). \tag{3.7}$$

3.3 Fiber Propagation as Local Unitary Evolution

Each photon propagates through its own optical fiber. At the level of the polarization qubit, the action of each fiber is described by a linear operator acting locally on the corresponding single-photon Hilbert space. In the ideal lossless description, this operator is unitary.

If U_A and U_B denote the single-qubit operators associated with the two fibers, the evolution of an initial two-photon state is

$$|\psi_{\mathrm{out}}\rangle = (U_A \otimes U_B)|\psi_{\mathrm{in}}\rangle. \tag{3.8}$$

This expression follows directly from the tensor-product structure of composite quantum systems and represents the natural bipartite extension of single-qubit evolution. It provides the fundamental starting point for subsequent refinements, including random unitary perturbations, concatenated fiber sections, and effective channel descriptions.

In this framework, the two fibers act independently on the two polarization subsystems, and the total evolution is obtained by tensor composition of the corresponding local transformations. This description remains valid as long as the fibers do not induce any direct interaction between the two photons.

The local-unitary model is therefore the first dynamical layer of the work: it captures deterministic polarization transformations while preserving the purity of the global two-photon state.

3.4 Pure-State and Density-Operator Descriptions

The first modeling level describes the two-photon system as a pure state in the bipartite Hilbert space $\mathbb{C}^2 \otimes \mathbb{C}^2$.

As introduced in Eq. (3.8), the evolution remains fully coherent, with the quantum state being represented directly through a state vector $|\psi\rangle$.

A more general and operationally complete description is obtained through the density operator formalism. For a pure bipartite state $|\psi\rangle$, the associated density operator is defined as

$$\rho_{AB} = |\psi\rangle\langle\psi|. \tag{3.9}$$

The density operator ρ_{AB} contains the complete statistical and physical information associated with the quantum state. In the pure-state regime, Eq. (3.9) is fully equivalent to the vector representation, since both descriptions encode the same physical state.

The density-operator formalism, however, extends naturally beyond pure states and therefore provides the appropriate framework for the subsequent modeling levels developed throughout this work. In particular, it becomes necessary when introducing:

- statistical ensembles of quantum states,
- reduced subsystem descriptions obtained through partial traces,
- effective non-unitary evolution induced by noisy channels,
- mixed states arising from decoherence and depolarization.

For this reason, although the initial propagation models can be formulated entirely at the pure-state level, the density-operator representation will progressively become central in the mathematical framework and in the quantitative analyses developed in the following chapters.

3.4.1 Expectation Values

In the pure-state formalism, the expectation value of an observable O is

$$\langle O\rangle = \langle\psi|O|\psi\rangle.$$

In the density-operator formalism, this expression generalizes to

$$\langle O \rangle = \mathrm{Tr}(\rho O). \tag{3.10}$$

This formula applies both to pure states and to mixed states. Indeed, for $\rho = |\psi\rangle\langle\psi|$, one recovers

$$\mathrm{Tr}(\rho O) = \mathrm{Tr}\left(|\psi\rangle\langle\psi|O\right) = \langle\psi|O|\psi\rangle.$$

This unified expression is used throughout the simulator to compute local quantities, state metrics, and two-photon observables from the same matrix-based representation.

3.4.2 Reduced Density Operators

A central operation in the density-operator formalism is the partial trace. It allows to describe the state accessible to a single subsystem when the global bipartite system is described by ρ_{AB}.

The reduced density operators are defined as

$$\rho_A = \mathrm{Tr}_B(\rho_{AB}), \qquad \rho_B = \mathrm{Tr}_A(\rho_{AB}). \tag{3.11}$$

These operators contain all information accessible through local measurements on the corresponding subsystem. For any observable O_A acting only on arm A,

$$\langle O_A \rangle = \mathrm{Tr}\left[\rho_{AB}\left(O_A \otimes I_B\right)\right] = \mathrm{Tr}(\rho_A O_A). \tag{3.12}$$

An explicit expression for the partial trace over subsystem B is

$$\rho_A = \sum_{j=0}^{1} \left(I_A \otimes \langle j|\right) \rho_{AB} \left(I_A \otimes |j\rangle\right), \tag{3.13}$$

where $\{|0\rangle, |1\rangle\}$ is the computational basis of subsystem B. The analogous expression defines ρ_B by tracing over subsystem A.

For the Bell state

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle + |10\rangle\right),$$

the reduced states are maximally mixed:

$$\rho_A = \rho_B = \frac{1}{2}I. \tag{3.14}$$

The operator $I/2$ represents the maximally mixed single-qubit state. In this situation, the two computational-basis outcomes occur with equal probability, and no preferred polarization direction exists locally.

Consequently, each photon individually is described by a state of maximal local uncertainty, even though the global bipartite state remains pure and maximally entangled. This result illustrates one of the fundamental properties of entanglement: complete information about the quantum state is encoded nonlocally in the correlations between the two subsystems rather than in the local states themselves.

3.4.3 Extension to Mixed-State Descriptions

The density-operator formalism also describes states that cannot be represented by a single state vector. This situation appears when different realizations of the system are combined statistically, or when unresolved degrees of freedom are not modeled explicitly.

A generic mixed state can be written as a convex combination

$$\rho = \sum_k p_k |\psi_k\rangle\langle\psi_k|, \qquad p_k \geq 0, \qquad \sum_k p_k = 1. \tag{3.15}$$

In the context of fiber propagation, this representation is useful for describing random phase fluctuations. If each photon pair experiences a different phase realization θ_k, the ensemble state is

$$\rho_{AB}^{(\mathrm{ens})} = \sum_k p_k |\psi(\theta_k)\rangle\langle\psi(\theta_k)|. \tag{3.16}$$

For a uniform numerical average over N sampled realizations, this becomes

$$\rho_{AB}^{(\mathrm{ens})} = \frac{1}{N} \sum_{k=1}^{N} |\psi(\theta_k)\rangle\langle\psi(\theta_k)|. \tag{3.17}$$

The density-operator formalism is therefore the natural framework for describing statistical phase averaging, local reduced states, and later effective non-unitary dynamics through quantum channels.

3.4.4 Bloch-Sphere Interpretation of Mixed States

For a single qubit, any density operator can be represented in Bloch form as

$$\rho = \frac{1}{2}\left(I + \mathbf{r}\cdot\boldsymbol{\sigma}\right), \tag{3.18}$$

where $\mathbf{r} \in \mathbb{R}^3$ is the Bloch vector and

$$\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z).$$

Physical states satisfy

$$|\mathbf{r}| \leq 1. \tag{3.19}$$

Pure states lie on the surface of the Bloch sphere and satisfy $|\mathbf{r}| = 1$. Mixed states satisfy $|\mathbf{r}| < 1$ and are represented by points inside the Bloch ball. The maximally mixed state corresponds to

$$\mathbf{r} = 0, \qquad \rho = \frac{1}{2} I,$$

which is the center of the Bloch representation.

Within this geometric picture, single-qubit unitary evolution acts as a rotation of the Bloch vector and preserves its norm. Statistical averaging and depolarizing processes instead reduce the magnitude of the Bloch vector, producing a contraction toward the center of the Bloch ball.

This distinction is useful for interpreting the models developed in this work. Coherent phase evolution changes the orientation of the state or of the measurement correlations without reducing purity. Statistical averaging and depolarization instead produce effective mixedness, visible as a loss of coherence or polarization information.

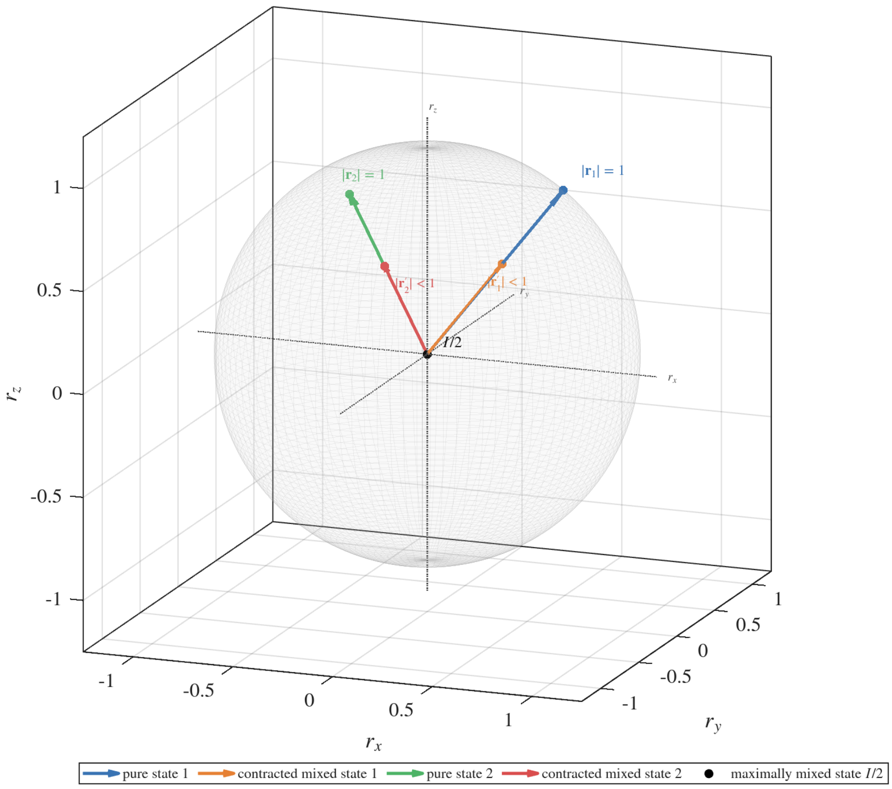


Figure 3.1: Bloch-sphere representation of pure and mixed single-qubit states. Pure states lie on the sphere surface and satisfy $|\mathbf{r}| = 1$, while mixed states are represented by contracted Bloch vectors with $|\mathbf{r}| < 1$. Statistical averaging and depolarizing mechanisms correspond to contractions toward the center of the Bloch ball.

3.5 Reduced Phase Model

Although a general fiber transformation may involve arbitrary polarization rotations, birefringence, and mode-coupling effects, the first approximation adopted here is simpler and directly motivated by the experimental control scheme.
After partial compensation of polarization drifts, the dominant residual effect is modeled as a relative phase shift between the horizontal and vertical polarization components. In the qubit representation, this corresponds to an effective rotation around the computational Z-axis of the Bloch sphere.

At the single-qubit level, this is represented by the diagonal unitary

$$U(\theta) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix}, \tag{3.20}$$

defined up to an irrelevant global phase.

If the two fibers induce phases θ_A and θ_B, the total transformation is

$$U_A \otimes U_B = U(\theta_A) \otimes U(\theta_B). \tag{3.21}$$

Applying this operator to the reference Bell state $|\Psi^+\rangle$ yields

$$|\psi(\theta_A, \theta_B)\rangle = \frac{1}{\sqrt{2}} \left(e^{i\theta_B}|01\rangle + e^{i\theta_A}|10\rangle \right). \tag{3.22}$$

The phase accumulated by each computational component depends on which subsystem carries the logical state $|1\rangle$. In particular, the component $|01\rangle$ acquires the phase generated by subsystem B, while the component $|10\rangle$ acquires the phase generated by subsystem A.

Factoring out the global phase $e^{i\theta_B}$, which has no observable effect, gives the reduced one-parameter state

$$|\psi(\theta)\rangle = \frac{1}{\sqrt{2}} \left(|01\rangle + e^{i\theta}|10\rangle \right), \qquad \theta = \theta_A - \theta_B. \tag{3.23}$$

Consequently, only the relative phase difference between the two arms has physical significance. Any common phase contribution acts as a global phase factor and therefore does not modify observable quantities.

With the convention adopted throughout this work, the reduced state in Eq. (3.23) can be interpreted equivalently as arising from a local phase rotation acting only on subsystem A,

$$U_A(\theta) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix}, \qquad U_B = I, \tag{3.24}$$

so that

$$(U_A \otimes I)|\Psi^+\rangle = \frac{1}{\sqrt{2}} \left(|01\rangle + e^{i\theta}|10\rangle \right). \tag{3.25}$$

This convention is adopted for consistency across the analytical derivations and numerical implementation developed throughout the present work.

Interpretation. In this reduced description, the physically relevant fiber action is encoded in a single parameter: the relative phase θ, determined by propagation conditions such as birefringence, path length, and environmental fluctuations.
The model isolates the dominant contribution affecting two-photon correlations while preserving a pure-state description. As a consequence, entanglement remains maximal and all observable variations arise from phase-induced modifications of quantum coherence.

Although a general fiber transformation may involve arbitrary polarization rotations, birefringence, and mode-coupling effects, the first approximation adopted here is simpler and directly motivated by the experimental control scheme.

Role of polarization compensation. Operationally, polarization compensation is used to realign the dominant output polarization basis with the measurement basis. For example, a known input polarization can be injected through the fiber and the compensating optical elements can be adjusted so that the detected output is again aligned with the expected horizontal–vertical reference basis. In this way, the largest polarization drifts are effectively suppressed at the level of the measurement setup.

This compensation, however, does not amount to a complete reconstruction of the full fiber transformation. In particular, it cannot generally remove all residual rotations around the stabilized polarization axis. After the dominant basis drift has been compensated, the remaining unresolved action is therefore modeled as a relative phase accumulation between the two orthogonal polarization components.

Consequently, the compensated fiber is described in this reduced model not as a generic uncontrolled single-qubit unitary, but as an effective phase operator acting on each arm. For arms A and B, this leads to the local transformation

$$U_A(\theta_A) \otimes U_B(\theta_B),$$

where θ_A and θ_B represent the residual phase shifts accumulated in the two paths.

Role in the modeling hierarchy. The reduced phase model constitutes the first nontrivial propagation model in the hierarchy. It provides an analytically tractable reference case for validating the simulator and serves as the baseline for subsequent extensions, including general unitary dynamics, ensemble descriptions, and effective non-unitary evolutions.

3.6 Correlation Tensor Representation

Bipartite correlations are described through the Pauli-operator basis

$$\{\sigma_i \otimes \sigma_j\}_{i,j\in\{x,y,z\}} .$$

constructed from the single-qubit Pauli matrices $\sigma_x, \sigma_y, \sigma_z$. It is also useful to introduce $\sigma_0 = I$, so that local terms and correlations can be treated within a single notation.

For a two-qubit state ρ_{AB}, the expectation values

$$T_{ij} = \mathrm{Tr}\left(\rho_{AB}\, \sigma_i \otimes \sigma_j\right), \qquad i, j \in \{x, y, z\}, \tag{3.26}$$

define the components of the correlation tensor $T \in \mathbb{R}^{3\times 3}$.

More generally, the state admits the Pauli decomposition

$$\rho_{AB} = \frac{1}{4} \sum_{\mu,\nu\in\{0,x,y,z\}} S_{\mu\nu}\, \sigma_\mu \otimes \sigma_\nu, \qquad S_{\mu\nu} = \mathrm{Tr}\left[\rho_{AB}\left(\sigma_\mu \otimes \sigma_\nu\right)\right], \tag{3.27}$$

where S_{ij} are generalized Stokes parameters. This representation is commonly used in quantum-state tomography of two-qubit photonic systems, where the density

operator is reconstructed from measured expectation values of tensor products of Pauli operators [2].
In this representation:

- S_{i0} and S_{0j} encode the local Bloch vectors of the two subsystems,
- the block S_{ij}, with $i, j \in \{x, y, z\}$, corresponds to the correlation tensor T_{ij}.

The orthogonality relation

$$\mathrm{Tr}\left[(\sigma_i \otimes \sigma_j)(\sigma_k \otimes \sigma_l)\right] = 4\,\delta_{ik}\delta_{jl} \tag{3.28}$$

ensures that all coefficients in Eq. (3.27) are obtained by direct projection.

Analytical behavior in the reduced phase model. For the phase-evolved Bell state in Eq. (3.23), the correlation tensor is

$$T(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{3.29}$$

This structure shows that the parameter θ induces a rotation in the transverse x-y correlation plane, while leaving the longitudinal z-component invariant.

The commonly evaluated diagonal correlations are

$$\begin{aligned} \langle \sigma_x \otimes \sigma_x \rangle &= T_{xx} = \cos\theta, \\ \langle \sigma_y \otimes \sigma_y \rangle &= T_{yy} = \cos\theta, \\ \langle \sigma_z \otimes \sigma_z \rangle &= T_{zz} = -1. \end{aligned} \tag{3.30}$$

Interpretation. The reduced phase model modifies only coherence terms, producing a rigid rotation of correlations in the transverse plane. Population correlations remain unchanged, reflecting the purely unitary nature of the transformation.
The correlation tensor is therefore the natural object for distinguishing changes in measurement alignment from actual loss of quantum coherence or entanglement.

Dependence on the local operator basis. The numerical values of the tensor components T_{ij} depend on the choice of local measurement axes used to define the Pauli operators.
Under local unitary transformations,

$$\rho_{AB} \;\to\; (U_A \otimes U_B)\,\rho_{AB}\,(U_A^\dagger \otimes U_B^\dagger),$$

the correlation tensor transforms according to a corresponding rotation of the local operator basis.
Consequently, individual tensor components are representation-dependent, while physically measurable quantities such as expectation values, entanglement measures, and Bell-inequality violations remain invariant under consistent changes of basis.
This distinction becomes particularly important when introducing arbitrary local measurement axes and optimized CHSH measurements in the following sections.

3.7 Arbitrary Local Measurement Axes

The correlation tensor introduced previously provides a complete description of all two-qubit correlations accessible through local Pauli measurements. This structure enables the prediction of measurement outcomes along arbitrary directions on the Bloch sphere.

A single-qubit observable associated with a unit vector

$$\mathbf{n} = \begin{pmatrix} n_x \\ n_y \\ n_z \end{pmatrix}, \qquad \|\mathbf{n}\| = 1,$$

is defined as

$$\sigma(\mathbf{n}) = \mathbf{n} \cdot \boldsymbol{\sigma} = n_x\sigma_x + n_y\sigma_y + n_z\sigma_z. \tag{3.31}$$

The use of a three-component vector to parametrize measurement directions follows from the structure of single-qubit observables. Any Hermitian operator acting on a two-dimensional Hilbert space can be decomposed as

$$O = \alpha I + \sum_{i\in\{x,y,z\}} n_i\sigma_i, \tag{3.32}$$

where $\alpha \in \mathbb{R}$ and σ_i are the Pauli matrices.

The identity term does not contribute to correlation measurements in the same way as the traceless Pauli part, since it produces only a uniform contribution. As a consequence, the physically relevant dichotomic observables can be represented by the traceless operator $\sigma(\mathbf{n})$.
The condition $\|\mathbf{n}\| = 1$ ensures that $\sigma(\mathbf{n})$ has eigenvalues ± 1, corresponding to a valid projective measurement with two possible outcomes.

For a bipartite state ρ_{AB}, choosing local directions $\mathbf{a}$ and $\mathbf{b}$ leads to the correlation function

$$E(\mathbf{a}, \mathbf{b}) = \mathrm{Tr}\left[\rho_{AB}\left(\sigma(\mathbf{a}) \otimes \sigma(\mathbf{b})\right)\right]. \tag{3.33}$$

Expanding in the Pauli basis gives

$$E(\mathbf{a}, \mathbf{b}) = \sum_{i,j\in\{x,y,z\}} a_i\, T_{ij}\, b_j, \tag{3.34}$$

or, in compact matrix notation,

$$E(\mathbf{a}, \mathbf{b}) = \mathbf{a}^{\mathsf{T}} T \mathbf{b}. \tag{3.35}$$

This expression shows that the correlation tensor completely determines all possible measurement outcomes for local Pauli observables.

Analytical behavior for the reduced phase model. Using Eq. (3.29), the correlation function for the phase-evolved Bell state becomes

$$E(\mathbf{a}, \mathbf{b}; \theta) = \mathbf{a}^{\mathsf{T}} T(\theta) \mathbf{b}. \tag{3.36}$$

For measurement directions restricted to the equatorial plane,

$$\mathbf{a} = \begin{pmatrix} \cos\alpha \\ \sin\alpha \\ 0 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} \cos\beta \\ \sin\beta \\ 0 \end{pmatrix},$$

this reduces to

$$E(\alpha, \beta; \theta) = \cos(\theta + \beta - \alpha). \tag{3.37}$$

Interpretation. The phase θ acts as an effective shift of the relative measurement angle. Consequently, phase evolution induced by the fiber is operationally equivalent to a rotation of measurement settings in the transverse plane.

Operational implications. In an experimental setting, arbitrary measurement axes are realized through local unitary transformations, such as wave plates, followed by a fixed computational-basis measurement.
This correspondence allows the theoretical model to reproduce realistic measurement procedures by combining local unitary rotations, standard projective measurements, and correlation evaluation through the tensor structure.

3.8 Quantum Channels and Depolarization

The ensemble description introduced in the previous section models statistical fluctuations by averaging over a family of unitary realizations. In that framework, each individual evolution remains coherent, and mixedness originates from classical uncertainty over the applied transformations.

More generally, however, a quantum system may interact with additional degrees of freedom that are not observed explicitly. In such cases, the reduced dynamics of the subsystem can no longer be described solely through unitary evolution acting on the state vector. Instead, the evolution must be formulated directly at the level of the density operator.

This motivates the introduction of quantum channels, which provide the general mathematical framework for effective non-unitary evolution.

Quantum channels. A quantum channel is a linear map

$$\mathcal{E} : \rho \mapsto \mathcal{E}(\rho), \tag{3.38}$$

acting on density operators and satisfying the conditions of complete positivity and trace preservation.

A general representation of a quantum channel is given by the operator-sum decomposition [1, 3]

$$\mathcal{E}(\rho) = \sum_{\mu} K_\mu \rho K_\mu^\dagger, \qquad \sum_{\mu} K_\mu^\dagger K_\mu = I, \tag{3.39}$$

where the operators K_μ are referred to as Kraus operators and encode the effective action of the physical process.

The channel formalism extends the unitary description introduced previously. While coherent propagation is represented through transformations of the form

$$\rho \to U\rho U^\dagger,$$

more general physical processes may produce irreversible loss of coherence and therefore require a non-unitary reduced description.

Global depolarizing channel. Among the simplest and most widely used quantum channels is the depolarizing channel, which models isotropic loss of coherence. In the two-qubit setting considered in this work, the global depolarizing model is written as

$$\mathcal{E}_p(\rho) = (1-p)\rho + \frac{p}{4} I_4, \tag{3.40}$$

where I_4 denotes the identity operator on the four-dimensional bipartite Hilbert space and p quantifies the strength of depolarization.

This channel continuously interpolates between coherent evolution and complete mixing, providing a simple benchmark model for the study of entanglement degradation and correlation suppression in bipartite systems.

Application to Bell states. Applying the channel to the Bell state introduced in Eq. (3.5),

$$\rho_0 = |\Psi^+\rangle\langle\Psi^+|,$$

produces the family of mixed states

$$\rho(p) = (1-p)\rho_0 + \frac{p}{4} I_4. \tag{3.41}$$

This construction provides a controlled framework for analyzing the degradation of bipartite quantum correlations under effective noisy evolution.

The global depolarizing model introduced here should be understood as a reference channel rather than as a complete physical model of two-arm fiber propagation. The refinement to independent local depolarizing channels acting on the two propagation arms is developed later in Chapter 5.

Chapter 4

Quantitative Characterization of Bipartite Quantum States

4.1 Reduced Density Operators as Local Diagnostics

Reduced density operators were introduced in Section 3.4.2 as the local states obtained from the global bipartite density operator by partial tracing over the complementary subsystem. In the present chapter, they are used as diagnostic tools for characterizing the local properties of each photon.

For a two-photon state ρ_{AB}, the reduced states are

$$\rho_A = \mathrm{Tr}_B(\rho_{AB}), \qquad \rho_B = \mathrm{Tr}_A(\rho_{AB}).$$

They determine all expectation values of observables acting locally on the corresponding subsystem. For instance, for an observable O_A acting only on arm A,

$$\langle O_A \rangle = \mathrm{Tr}(\rho_A O_A).$$

Thus, ρ_A and ρ_B represent the complete local information accessible through measurements performed independently on the two optical arms.

For Pauli observables, the local expectation values are

$$\begin{aligned} \langle \sigma_x \rangle_A &= \mathrm{Tr}(\rho_A \sigma_x), \\ \langle \sigma_y \rangle_A &= \mathrm{Tr}(\rho_A \sigma_y), \\ \langle \sigma_z \rangle_A &= \mathrm{Tr}(\rho_A \sigma_z). \end{aligned} \tag{4.1}$$

Analogous expressions hold for subsystem B. These quantities define the local Bloch-vector components of each reduced state and quantify the degree of local polarization.

More generally, a reduced density operator of the form

$$\rho = \frac{1}{d} I$$

represents a maximally mixed state in a d-dimensional Hilbert space. In this situation, all basis states occur with equal probability and no preferred measurement direction exists.

For single-qubit systems ($d = 2$), the maximally mixed state is therefore

$$\rho = \frac{1}{2} I.$$

Conversely, when

$$\rho \neq \frac{1}{2} I,$$

the subsystem possesses nontrivial local structure. In the Bloch-sphere representation, this corresponds to a nonzero Bloch vector and therefore to the existence of preferred local measurement outcomes or residual polarization.

For the Bell state

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}} \left(|01\rangle + |10\rangle\right),$$

one obtains

$$\rho_A = \rho_B = \frac{1}{2} I.$$

Consequently,

$$\langle \sigma_x \rangle_A = \langle \sigma_y \rangle_A = \langle \sigma_z \rangle_A = 0,$$

and similarly for subsystem B. Each photon is therefore locally unpolarized, even though the global two-photon state is pure and maximally entangled.

The same local behavior holds for the reduced phase model introduced in Section 3.5. The state

$$|\psi(\theta)\rangle = \frac{1}{\sqrt{2}} \left(|01\rangle + e^{i\theta}|10\rangle\right)$$

has reduced density operators

$$\rho_A(\theta) = \rho_B(\theta) = \frac{1}{2} I.$$

The relative phase θ changes the global coherence structure and the two-photon correlations, but it does not modify the local reduced states. Therefore, local polarization measurements alone cannot detect the fixed phase shift.

By linearity of the partial trace, the same conclusion extends to the random-phase ensemble introduced in Section 3.4.3. If

$$\rho_{AB}^{(\mathrm{ens})} = \sum_k p_k |\psi(\theta_k)\rangle\langle\psi(\theta_k)|, \qquad \sum_k p_k = 1,$$

then

$$\rho_A^{(\mathrm{ens})} = \rho_B^{(\mathrm{ens})} = \frac{1}{2} I.$$

Thus, in the reduced phase model and in its statistical extension, the local states remain maximally mixed. The effect of the phase distribution appears at the global level, through the decay of coherences and correlations, rather than through a change in local polarization.

This observation is important for the interpretation of entanglement. The same local density operator

$$\rho_A = \frac{1}{2}I$$

can arise from physically different situations. It may describe a classical statistical mixture,

$$\rho^{(\mathrm{class})} = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1|,$$

or it may arise as the reduced state of an entangled bipartite system,

$$\rho_A^{(\mathrm{ent})} = \mathrm{Tr}_B\left(|\Psi^+\rangle\langle\Psi^+|\right) = \frac{1}{2}I.$$

These two cases are indistinguishable by local measurements alone. Their difference is visible only at the global level, where one must analyze global purity, bipartite correlations, and entanglement measures.

For this reason, reduced density operators are not sufficient by themselves to characterize the full bipartite state. They provide local information, while quantities such as global purity, fidelity, concurrence, tangle, entanglement of formation, and CHSH parameters are required to quantify the global coherence, entanglement, and nonlocality of the two-photon system.

4.2 Purity and Mixedness

Having introduced reduced density operators in Section 4.1, we now require a quantitative criterion to distinguish whether a quantum state is pure or mixed.

A density operator represents a pure quantum state when it can be expressed in the form of Eq. (3.9), namely as the projector onto a single state vector.
Otherwise, the state is mixed, meaning that it describes a statistical ensemble of pure states with associated probabilities.

A convenient quantitative measure of mixedness is the purity, defined as

$$\gamma(\rho) = \mathrm{Tr}(\rho^2). \tag{4.2}$$

This quantity satisfies

$$\gamma(\rho) = 1 \iff \rho \text{ is pure}, \qquad \gamma(\rho) < 1 \iff \rho \text{ is mixed}. \tag{4.3}$$

If $\rho = \sum_i \lambda_i |i\rangle\langle i|$ is its spectral decomposition, then

$$\gamma(\rho) = \sum_i \lambda_i^2, \tag{4.4}$$

which shows that purity measures the concentration of the eigenvalue distribution. More generally, for a Hilbert space of dimension d,

$$\frac{1}{d} \leq \gamma(\rho) \leq 1, \tag{4.5}$$

with the lower bound attained by the maximally mixed state $\rho = I/d$.

For single-qubit systems, purity admits a simple geometric interpretation. As shown in 3.18, any density operator can be written as

$$\rho = \frac{1}{2}\left(I + \vec{r} \cdot \vec{\sigma}\right), \tag{4.6}$$

where $\vec{r} \in \mathbb{R}^3$ is the Bloch vector. One then finds

$$\gamma(\rho) = \frac{1 + \|\vec{r}\|^2}{2}. \tag{4.7}$$

Thus,

$$\|\vec{r}\| = 1 \Rightarrow \rho \text{ is pure}, \qquad \|\vec{r}\| = 0 \Rightarrow \rho = \frac{I}{2}.$$

Mixedness does not uniquely determine the physical origin of a state. As discussed in Section 4.1, it may arise either from classical statistical uncertainty or from entanglement with an unobserved subsystem.

A paradigmatic example is provided by the Bell state $|\Psi^+\rangle$, whose global density operator

$$\rho_{AB} = |\Psi^+\rangle\langle\Psi^+|$$

is pure,

$$\gamma(\rho_{AB}) = 1,$$

while the reduced states satisfy

$$\rho_A = \rho_B = \frac{I}{2}, \qquad \gamma(\rho_A) = \gamma(\rho_B) = \frac{1}{2}.$$

This illustrates that a globally pure entangled state may generate locally maximally mixed subsystems.

In the present model, the phase-evolved state introduced in Chapter 3,

$$|\psi(\theta)\rangle = \frac{|01\rangle + e^{i\theta}|10\rangle}{\sqrt{2}},$$

remains pure for all values of θ,

$$\gamma(\rho(\theta)) = 1, \tag{4.8}$$

while the reduced states remain maximally mixed,

$$\rho_A(\theta) = \rho_B(\theta) = \frac{I}{2}.$$

Thus, relative phase shifts modify correlation observables without affecting either global purity or local mixedness.

When extending the model to include stochastic or environmental effects, the situation changes. For the random-phase ensemble, the global state depends on the coherence parameter

$$\mu = \langle e^{i\theta} \rangle, \tag{4.9}$$

and its purity is

$$\gamma\left(\rho_{AB}^{(\mathrm{ens})}\right) = \frac{1 + |\mu|^2}{2} = \frac{1 + \langle \cos\theta \rangle^2 + \langle \sin\theta \rangle^2}{2}. \tag{4.10}$$

This shows that purity directly quantifies the loss of phase coherence in the ensemble: the state is pure for $|\mu| = 1$ and becomes mixed as $|\mu|$ decreases.

Purity therefore provides a fundamental diagnostic tool for assessing coherence loss in the simulator, complementing correlation observables and preparing the ground for fidelity and entanglement measures.

4.3 Fidelity with Respect to Bell States

Having introduced reduced density operators and purity as measures of local information and mixedness, we now require a quantitative tool to evaluate how close a given quantum state is to a reference target state. This role is fulfilled by the notion of fidelity.

In general, the fidelity between two quantum states described by density operators ρ and σ is defined as

$$F(\rho, \sigma) = \left(\mathrm{Tr} \sqrt{\sqrt{\rho}\, \sigma \sqrt{\rho}} \right)^2, \tag{4.11}$$

and satisfies

$$0 \leq F(\rho, \sigma) \leq 1,$$

with equality to 1 if and only if $\rho = \sigma$.

In the present context, the reference state is pure, $\sigma = |\phi\rangle\langle\phi|$, and the expression simplifies to

$$F(\rho, |\phi\rangle) = \langle \phi | \rho | \phi \rangle, \tag{4.12}$$

which admits a direct interpretation as the probability of projecting onto $|\phi\rangle$.

We focus on Bell state $|\Psi^+\rangle$ and define

$$F_{\Psi^+}(\rho) = \langle \Psi^+ | \rho | \Psi^+ \rangle. \tag{4.13}$$

For the phase-evolved state $|\psi(\theta)\rangle$, with $\rho(\theta) = |\psi(\theta)\rangle\langle\psi(\theta)|$, one obtains

$$F_{\Psi^+}(\rho(\theta)) = |\langle \Psi^+|\psi(\theta)\rangle|^2,$$

with

$$\langle \Psi^+|\psi(\theta)\rangle = \frac{1}{2}\left(1 + e^{i\theta}\right),$$

leading to

$$F_{\Psi^+}(\rho(\theta)) = \frac{1+\cos\theta}{2}. \tag{4.14}$$

Thus, the relative phase modifies the overlap with the reference Bell state, even though the global state remains pure. In particular,

$$F_{\Psi^+} = 1 \quad \text{for } \theta = 0, \qquad F_{\Psi^+} = 0 \quad \text{for } \theta = \pi.$$

For ensemble states,

$$\rho_{AB}^{(\mathrm{ens})} = \frac{1}{N}\sum_{k=1}^{N} |\psi(\theta_k)\rangle\langle\psi(\theta_k)|,$$

linearity yields

$$F_{\Psi^+}\left(\rho_{AB}^{(\mathrm{ens})}\right) = \frac{1}{N}\sum_{k=1}^{N} F_{\Psi^+}\left(\rho(\theta_k)\right).$$

Introducing the coherence parameter $\mu = \langle e^{i\theta}\rangle$, one finds

$$F_{\Psi^+} = \frac{1}{2}\left(1 + \mathrm{Re}(\mu)\right) = \frac{1}{2}\left(1 + \langle\cos\theta\rangle\right). \tag{4.15}$$

In particular, for a uniform distribution over $[0, 2\pi]$,

$$F_{\Psi^+} = \frac{1}{2}.$$

Fidelity therefore provides a quantitative measure of how closely a given state approximates the target Bell state. Unlike purity, which characterizes mixedness, fidelity is sensitive to coherent deformations of the state and complements correlation observables in the analysis of the system.

4.4 Concurrence as an Entanglement Measure

Reduced density operators, purity, and fidelity provide complementary information about a bipartite quantum state, but none of them directly quantifies entanglement. In particular, fidelity with respect to a Bell state may vary under local unitary evolution even when the amount of entanglement remains unchanged. A genuine entanglement measure is therefore required.

For two-qubit systems, a standard measure is the concurrence, defined such that

$$0 \leq C \leq 1,$$

where $C = 0$ corresponds to separable states and $C = 1$ to maximally entangled states.

For a pure two-qubit state

$$|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle,$$

the concurrence is

$$C(|\psi\rangle) = 2|ad - bc|. \tag{4.16}$$

This expression vanishes for separable states and reaches unity for maximally-entangled states.

For Bell state:

$$|\Psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}} \quad \Rightarrow \quad C(|\Psi^+\rangle) = 1.$$

For the phase-evolved state:

$$|\psi(\theta)\rangle = \frac{|01\rangle + e^{i\theta}|10\rangle}{\sqrt{2}}, \quad \Rightarrow \quad C(|\psi(\theta)\rangle) = 1, \tag{4.17}$$

showing that the relative phase modifies fidelity but leaves entanglement unchanged. This reflects the invariance of bipartite entanglement under local unitary transformations.

For pure bipartite states, concurrence is directly related to the reduced density operators:

$$\begin{aligned} C(|\psi\rangle) &= 2\sqrt{\det(\rho_A)} = \sqrt{2\,(1 - \gamma(\rho_A))} \\ &= 2\sqrt{\det(\rho_B)} = \sqrt{2\,(1 - \gamma(\rho_B))}, \end{aligned} \tag{4.18}$$

where

$$\rho_A = \mathrm{Tr}_B\left(|\psi\rangle\langle\psi|\right), \qquad \rho_B = \mathrm{Tr}_A\left(|\psi\rangle\langle\psi|\right).$$

Thus, entanglement is encoded in the mixedness of the reduced states: separable states yield pure reductions, while maximally entangled states yield maximally mixed reductions.

For mixed states $\rho \in \mathbb{C}^{4\times 4}$, concurrence is defined through the Wootters spin-flip construction [4]. The associated spin-flipped density operator is

$$\tilde{\rho} = (\sigma_y \otimes \sigma_y)\rho^*(\sigma_y \otimes \sigma_y). \tag{4.19}$$

Let $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \lambda_4 \geq 0$ be the square roots of the eigenvalues of $\rho\tilde{\rho}$. Then

$$C(\rho) = \max\left(0, \lambda_1 - \lambda_2 - \lambda_3 - \lambda_4\right). \tag{4.20}$$

This expression applies to arbitrary two-qubit states and reduces to the pure-state formula in the appropriate limit.

In the present model, mixed states arise from ensemble averaging over random phases,

$$\rho_{AB}^{(\mathrm{ens})} = \frac{1}{N} \sum_{k=1}^{N} |\psi(\theta_k)\rangle\langle\psi(\theta_k)|,$$

which depend on the coherence parameter $\mu = \langle e^{i\theta} \rangle$.

In the computational basis, the ensemble state reads

$$\rho_{AB}^{(\mathrm{ens})} = \frac{1}{2}|01\rangle\langle 01| + \frac{1}{2}|10\rangle\langle 10| + \frac{\mu}{2}|10\rangle\langle 01| + \frac{\mu^*}{2}|01\rangle\langle 10|. \tag{4.21}$$

For this family of states, the concurrence is

$$C\left(\rho_{AB}^{(\mathrm{ens})}\right) = |\mu|. \tag{4.22}$$

This result shows that the same coherence parameter controlling purity and fidelity also governs entanglement degradation:

$$\gamma\left(\rho_{AB}^{(\mathrm{ens})}\right) = \frac{1 + |\mu|^2}{2}, \qquad F_{\Psi^+} = \frac{1 + \mathrm{Re}(\mu)}{2}.$$

Thus, the reduced phase model provides a unified description in which coherence, mixedness, and entanglement are all controlled by a single parameter.

4.5 Tangle

While concurrence provides a direct quantitative measure of bipartite entanglement, it is often useful to introduce an associated quadratic quantity known as the tangle. For a two-qubit state ρ, the tangle is defined as

$$\tau(\rho) = C(\rho)^2, \tag{4.23}$$

where $C(\rho)$ denotes the concurrence.

Since concurrence satisfies

$$0 \leq C(\rho) \leq 1,$$

the tangle also satisfies

$$0 \leq \tau(\rho) \leq 1.$$

The tangle therefore vanishes for separable states and reaches unity for maximally entangled Bell states.

For pure bipartite states, the tangle admits an equivalent representation in terms of the reduced density operator. If

$$|\psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B,$$

then

$$\tau = 4 \det(\rho_A), \tag{4.24}$$

where

$$\rho_A = \mathrm{Tr}_B \left(|\psi\rangle\langle\psi| \right).$$

This expression shows directly that the tangle quantifies the degree of non-purity of the reduced subsystem induced by bipartite entanglement.

For the Bell state $|\Psi^+\rangle$, one obtains

$$C = 1, \qquad \tau = 1,$$

while for separable product states,

$$C = 0, \qquad \tau = 0.$$

Compared to concurrence, the quadratic dependence of the tangle emphasizes the suppression of weak entanglement contributions. Consequently, the tangle often exhibits a sharper degradation under noisy evolution.

The tangle also plays an important role in the study of multipartite entanglement sharing and monogamy relations, where squared concurrence naturally appears in entanglement-balance inequalities.

Within the present work, the tangle is used as an auxiliary entanglement metric complementing concurrence and providing an alternative characterization of entanglement degradation under effective fiber propagation.

4.6 Entanglement of Formation

The concurrence and tangle quantify the degree of bipartite entanglement present in a quantum state. In particular, larger values of these quantities correspond to stronger nonclassical correlations between the two subsystems.

However, these measures do not directly characterize the physical resources required to generate such entanglement. In quantum information theory, entanglement is treated as a resource, and one may ask how much pure-state entanglement is necessary in order to prepare a given mixed state.

For this purpose, a central quantity is the entanglement of formation.

The entanglement of formation measures the minimal amount of bipartite entanglement required, on average, to prepare a given mixed state using pure entangled states together with local operations and classical communication.

For arbitrary quantum systems, the computation of entanglement of formation is generally difficult. Remarkably, however, Wootters showed that the entanglement of formation of arbitrary two-qubit states can be expressed analytically in terms of concurrence [4].

For a two-qubit density operator ρ, the entanglement of formation is defined as

$$E_F(\rho) = h\left(\frac{1+\sqrt{1-C(\rho)^2}}{2}\right), \tag{4.25}$$

where $C(\rho)$ denotes the concurrence and

$$h(x) = -x\log_2 x - (1-x)\log_2(1-x) \tag{4.26}$$

is the binary entropy function.

Since

$$0 \le C(\rho) \le 1,$$

the entanglement of formation satisfies

$$0 \le E_F(\rho) \le 1.$$

In particular:

$$E_F = 0$$

for separable states, while

$$E_F = 1$$

for maximally entangled Bell states.

Unlike concurrence, the entanglement of formation depends nonlinearly on the entanglement content of the state. As a consequence, small reductions in concurrence may produce strongly nonlinear reductions in E_F.

This nonlinear behavior is physically significant because it reflects the effective loss of entanglement resources available for quantum-information tasks such as teleportation, entanglement distribution, and quantum communication protocols.

Within the present work, the entanglement of formation provides an operational complement to concurrence and tangle, allowing the degradation of entanglement under effective fiber propagation to be interpreted not only geometrically, but also in terms of entanglement-resource loss.

4.7 Bell Inequalities and CHSH Nonlocality

The correlation structure introduced in the previous sections provides a complete operational description of measurement outcomes for arbitrary local observables. This framework makes it possible to formulate quantitative criteria that distinguish correlations compatible with classical physics from those that are intrinsically quantum.

The central question addressed by Bell inequalities is the following: *can the observed correlations between spatially separated subsystems be explained by a classical model based on shared hidden information?*

In a classical description based on local hidden variables, it is assumed that the outcomes of measurements performed on subsystems A and B are determined by a shared underlying variable λ, drawn from a probability distribution $p(\lambda)$. This variable represents all pre-existing information that could influence the measurement results.

For each realization of λ, the outcome of a measurement on subsystem A along direction $\mathbf{a}$ is described by a deterministic response function

$$A(\mathbf{a}, \lambda) \in \{-1, +1\},$$

and similarly for subsystem B,

$$B(\mathbf{b}, \lambda) \in \{-1, +1\}.$$

The key physical assumption is locality: the outcome at each subsystem depends only on the local measurement setting and on the shared variable λ, but not on the measurement choice at the distant subsystem.
Since λ is not directly accessible, observable quantities correspond to averages over its distribution. The classical correlation function is therefore defined as

$$E(\mathbf{a}, \mathbf{b}) = \int d\lambda\, p(\lambda)\, A(\mathbf{a}, \lambda)\, B(\mathbf{b}, \lambda), \qquad p(\lambda) \geq 0, \qquad \int d\lambda\, p(\lambda) = 1. \tag{4.27}$$

following the local hidden-variable framework introduced by Bell [5].

This expression represents the most general form of bipartite correlations compatible with a local hidden-variable model.

The most widely used form of a Bell inequality is the Clauser–Horne–Shimony–Holt (CHSH) inequality. It involves four measurement settings:

$$\mathbf{a}_1, \mathbf{a}_2 \quad \text{for subsystem } A, \qquad \mathbf{b}_1, \mathbf{b}_2 \quad \text{for subsystem } B.$$

The CHSH parameter is defined as

$$S = E(\mathbf{a}_1, \mathbf{b}_1) + E(\mathbf{a}_1, \mathbf{b}_2) + E(\mathbf{a}_2, \mathbf{b}_1) - E(\mathbf{a}_2, \mathbf{b}_2). \tag{4.28}$$

Classical bound. Under the assumptions of locality and realism, the CHSH parameter satisfies

$$|S| \leq 2. \tag{4.29}$$

This bound reflects a fundamental limitation of classical correlations: although each term in S can individually reach its maximal value, their combination is constrained by the requirement that all outcomes originate from the same underlying variable λ.

Quantum prediction. In quantum mechanics, correlations arise from the structure of the joint state rather than from pre-existing local variables. The correlation function is given by Eq. (3.33).
Due to the non-commuting nature of quantum observables and the presence of entanglement, the CHSH parameter can exceed the classical bound. The maximal value allowed by quantum mechanics is

$$|S| \leq 2\sqrt{2}, \tag{4.30}$$

known as the Tsirelson bound.
This violation does not imply faster-than-light communication. Rather, it implies that the correlations cannot be reproduced by any model based on local hidden variables.

Tensor formulation. Using the correlation tensor representation, the CHSH parameter can be expressed as

$$S = \mathbf{a}_1^\mathsf{T} T \mathbf{b}_1 + \mathbf{a}_1^\mathsf{T} T \mathbf{b}_2 + \mathbf{a}_2^\mathsf{T} T \mathbf{b}_1 - \mathbf{a}_2^\mathsf{T} T \mathbf{b}_2. \tag{4.31}$$

This formulation makes explicit that the CHSH inequality depends only on the correlation tensor and on the choice of measurement directions. In particular, it shows that the observed value of S is sensitive not only to the state, but also to the alignment between the measurement axes and the principal directions of the tensor.

A particularly important result is that the maximal violation achievable for a given state can be computed directly from the tensor. Defining

$$U = T^\mathsf{T} T, \tag{4.32}$$

and denoting by λ_1 and λ_2 the two largest eigenvalues of U, the maximal CHSH value is

$$S_{\max} = 2\sqrt{\lambda_1 + \lambda_2}. \tag{4.33}$$

Interpretation. A violation of the CHSH inequality, i.e. $|S| > 2$, indicates the presence of correlations that cannot be explained by any local hidden-variable model. Such correlations are referred to as nonlocal.

Nonlocality should be understood as a property of correlations rather than of individual subsystems: each subsystem alone may appear completely random, while the joint statistics reveal strong structured correlations.

It is important to distinguish nonlocality from entanglement. While all maximally entangled pure states achieve maximal quantum violation for suitable measurement axes, not all entangled states violate the CHSH inequality. In particular, sufficiently mixed states may remain within the classical bound despite being entangled.

Operational implications. The CHSH parameter provides a direct bridge between the theoretical description of correlations and experimentally accessible quantities. In practice, it is evaluated by performing measurements along four selected pairs of local axes and combining the resulting correlation values.

The correlation tensor entering this construction can be experimentally reconstructed from a set of bipartite expectation values in the Pauli basis, in a procedure closely related to quantum state tomography. In this sense, the evaluation of CHSH inequalities can be viewed as a specific post-processing of tomographically accessible data.

Two complementary perspectives are particularly relevant:

- evaluation of S for fixed measurement settings, corresponding to a specific experimental configuration,
- computation of $S_{\max}$, which quantifies the maximal nonlocality extractable from the state.

This separation will play a central role in the analysis of fiber-induced effects, where transformations may alter the observed correlations without necessarily reducing the underlying nonlocality.

4.8 Coherence, Entanglement, and Nonlocality in the Random-Phase Model

The previous sections introduced several complementary quantities used to characterize bipartite quantum states:

- purity, quantifying mixedness;
- fidelity, quantifying overlap with a reference state;
- concurrence and tangle, quantifying bipartite entanglement;
- entanglement of formation, quantifying entanglement resources;
- CHSH nonlocality, quantifying violation of local hidden-variable models.

Although these quantities are related, they characterize fundamentally different physical properties of the state.

Purity measures how close the density operator is to a pure state, independently of whether the state is entangled or separable.
Fidelity measures proximity to a chosen target state and therefore depends explicitly on the selected reference.
Concurrence and tangle quantify bipartite entanglement, while the entanglement of formation provides an operational interpretation of the entanglement content in terms of preparation resources.
Finally, CHSH inequalities probe nonlocal correlations that cannot be reproduced by any local hidden-variable model.

These distinctions are particularly important because entanglement and nonlocality are not equivalent concepts.

All pure maximally entangled Bell states violate the CHSH inequality maximally for suitable measurement settings. However, in general:

$$\text{entanglement} \neq \text{nonlocality}.$$

In particular, sufficiently mixed entangled states may fail to violate any CHSH inequality despite possessing nonzero concurrence and nonzero entanglement of formation.

The random-phase ensemble considered in this work provides a particularly useful example because the state properties are governed by a single coherence parameter

$$\mu = \langle e^{i\theta} \rangle = \langle \cos\theta \rangle + i \langle \sin\theta \rangle. \tag{4.34}$$

For this family of states, the relevant quantities become

$$\gamma(\rho) = \frac{1 + |\mu|^2}{2}, \tag{4.35}$$

$$C(\rho) = |\mu|, \qquad \tau(\rho) = |\mu|^2, \tag{4.36}$$

$$F_{\Psi^+} = \frac{1 + \mathrm{Re}(\mu)}{2}, \tag{4.37}$$

and

$$E_F(\rho) = h\left(\frac{1 + \sqrt{1 - |\mu|^2}}{2}\right), \tag{4.38}$$

where $h(x)$ denotes the binary entropy function introduced previously.

These expressions reveal the different sensitivities of the various quantities.

- Purity depends quadratically on $|\mu|$.
- Concurrence depends linearly on $|\mu|$.
- Tangle depends quadratically on $|\mu|$.
- Entanglement of formation depends nonlinearly on $|\mu|$.
- Fidelity depends only on the real part $\mathrm{Re}(\mu)$.

Consequently, two states may possess the same concurrence while having different fidelities, or may possess nonzero entanglement while exhibiting no observable CHSH violation.

This distinction becomes particularly evident in the fixed-phase model. In that case,

$$|\mu| = 1$$

for all phase values θ, because the evolution remains purely unitary. Therefore,

$$C = 1, \qquad \tau = 1, \qquad E_F = 1,$$

while the fidelity varies continuously with the phase:

$$F_{\Psi^+} = \frac{1 + \cos\theta}{2}.$$

Thus, changes in fidelity do not necessarily correspond to degradation of entanglement.

By contrast, ensemble averaging reduces the magnitude $|\mu|$, producing genuine decoherence and entanglement degradation. In this regime:

- the global purity decreases;
- concurrence decreases;
- tangle decreases more rapidly due to its quadratic dependence;
- entanglement of formation decreases nonlinearly;
- CHSH nonlocality may eventually disappear.

A useful geometric interpretation is obtained by representing μ in the complex plane.

- The concurrence is proportional to the distance of μ from the origin.
- The tangle depends on the squared distance.
- The fidelity corresponds to the projection onto the real axis.

Deterministic phase evolution moves μ along the unit circle without changing its magnitude, while ensemble averaging contracts μ toward the origin.

The limiting regimes illustrate these behaviors clearly:

$$|\mu| = 1 \Rightarrow \gamma(\rho) = 1, \quad C = 1, \quad \tau = 1, \quad E_F = 1, \tag{4.39}$$

while

$$|\mu| = 0 \Rightarrow \gamma(\rho) = \frac{1}{2}, \quad C = 0, \quad \tau = 0, \quad E_F = 0. \tag{4.40}$$

Intermediate values $0 < |\mu| < 1$ correspond to partially coherent mixed states with nonzero but degraded entanglement.

These relations are specific to the random-phase ensemble model considered in this work. For general two-qubit states, purity, fidelity, concurrence, entanglement of formation, and CHSH nonlocality are independent quantities and cannot generally be reduced to a single parameter.

In summary, the coherence parameter μ provides a unified description of the states analyzed throughout the present work. Its magnitude controls mixedness and entanglement, while its phase determines the orientation of observable correlations and the overlap with the reference Bell state.

The comparison between these quantities highlights an important conceptual hierarchy:

$$\text{mixedness} \quad \longrightarrow \quad \text{entanglement} \quad \longrightarrow \quad \text{nonlocality}.$$

This hierarchy will play a central role in the interpretation of the numerical experiments and in the analysis of effective fiber-induced degradation mechanisms.

4.9 Computational Evaluation of State Metrics

The quantities introduced throughout this chapter constitute the main diagnostic layer of the simulator. Their purpose is to translate the density-operator description of bipartite quantum states into quantitative indicators of mixedness, coherence, entanglement, and nonlocality.

At computational level, all metrics are evaluated directly from the numerically reconstructed density operator ρ_{AB}. This provides a unified characterization framework applicable to coherent phase evolution, statistical phase ensembles, depolarizing channels, and composite effective fiber models.

Reduced density operators are computed through partial-trace operations acting on the global bipartite state. The resulting local states are then used to evaluate local observables, local purities, and entanglement-related quantities.

The simulator evaluates purity, fidelity, concurrence, tangle, entanglement of formation, and CHSH quantities according to the definitions introduced in the previous sections. Since all quantities are derived from the same density-operator representation, the computational framework remains internally consistent across all propagation models and numerical experiments.

The simultaneous evaluation of multiple metrics is essential because different quantities characterize different physical properties of the propagated quantum state.
In particular:

- purity diagnoses mixedness and coherence loss,
- fidelity measures proximity to the reference Bell state,
- concurrence and tangle quantify bipartite entanglement,
- entanglement of formation quantifies entanglement resources,
- CHSH quantities characterize observable nonlocality.

This distinction is important for the interpretation of fiber-induced effects. For example, a reduction in fidelity may arise from coherent phase evolution without necessarily reducing entanglement, whereas reductions in concurrence, tangle, and entanglement of formation indicate degradation of quantum correlations.
Similarly, nonzero concurrence does not necessarily imply CHSH violation, particularly for sufficiently mixed states.

The coexistence of these complementary quantities therefore provides a multidimensional characterization of the propagation regime, allowing coherent phase evolution, statistical decoherence, entanglement degradation, and nonlocality suppression to be distinguished quantitatively throughout the simulator analysis.

Chapter 5

Effective Fiber Models

5.1 Motivation for Effective Fiber Models

The previous chapters established the mathematical framework required to describe bipartite polarization states, local unitary propagation, correlation measurements, quantum channels, and quantitative state metrics. Within that framework, fiber propagation was initially modeled through analytically controlled transformations, including coherent phase shifts, statistical phase ensembles, and idealized depolarizing maps.

These models provide a necessary validation baseline, but they do not yet fully capture the physical structure of a fiber-based quantum communication link. In the experimental scenario considered in this work, the two photons propagate through two distinct optical arms. Each arm acts locally on the polarization degree of freedom of the corresponding photon and may introduce both coherent and incoherent perturbations. Consequently, a physically meaningful fiber model must preserve the bipartite structure of the system while allowing independent evolution and degradation mechanisms on subsystems A and B.

The objective of the present chapter is therefore to construct effective fiber models connecting the abstract quantum-channel formalism introduced previously with the physical interpretation of entangled-photon propagation in optical fibers. The term "effective" is used deliberately: the goal is not to reproduce the full electromagnetic propagation problem inside the fiber, including all microscopic temporal and spectral degrees of freedom, but rather to develop reduced models describing the observable action of the fiber on the polarization state.

From coherent propagation to effective noisy evolution. In the ideal coherent regime, each fiber arm is represented by a local unitary transformation acting on the corresponding polarization qubit. The bipartite state therefore evolves according to

$$\rho_{AB}^{\text{out}} = (U_A \otimes U_B)\, \rho_{AB}^{\text{in}} \left(U_A^\dagger \otimes U_B^\dagger\right). \tag{5.1}$$

In this limit, the evolution preserves purity and entanglement, and the fiber acts as a coherent optical element producing deterministic polarization rotations and relative phase shifts.

Real optical fibers, however, may introduce additional propagation effects that cannot be represented solely through fixed polarization rotations. Unresolved temporal, spectral, and environmental fluctuations may effectively reduce polarization coherence when only polarization observables are retained in the description. At the level of the reduced polarization state, these effects appear as non-unitary dynamics.

Within the density-operator formalism introduced in Section 3.8, such mechanisms are described through effective local quantum channels acting independently on the two subsystems:

$$\rho_{AB}^{\text{out}} = (\mathcal{E}_A \otimes \mathcal{E}_B)\left(\rho_{AB}^{\text{in}}\right), \tag{5.2}$$

where $\mathcal{E}_A$ and $\mathcal{E}_B$ describe the effective action of the two fiber arms.

Entanglement sensitivity to local degradation. Entanglement is encoded in the nonlocal correlations of the joint bipartite state ρ_{AB}, rather than in the properties of either subsystem separately. Consequently, even moderate local perturbations acting on a single propagation arm may significantly degrade global quantum correlations.

This aspect is particularly relevant for quantum communication systems because many protocols rely directly on the preservation of bipartite entanglement. Bell-state fidelity, concurrence, correlation tensors, and CHSH nonlocality may all degrade even when local polarization statistics remain apparently reasonable.

For this reason, the physically relevant object is the full bipartite density operator together with its associated correlation and entanglement structure, rather than the reduced states ρ_A and ρ_B alone.

Modeling strategy of the present chapter. The construction developed in this chapter proceeds progressively. First, local depolarizing channels are introduced as effective models of independent polarization degradation in the two fiber arms. These channels are then combined with the coherent phase-evolution model developed previously, leading to composite effective fiber channels incorporating both residual unitary evolution and incoherent depolarization.

This progression establishes the bridge between the analytically controlled models introduced previously and the more physically meaningful effective propagation models studied later in the numerical experiments.

5.2 Local Depolarizing Fiber Channels

The global depolarizing model introduced in Section 3.8 provides a useful baseline for studying the degradation of bipartite quantum correlations. However, that model acts collectively on the full two-qubit state and does not explicitly distinguish the physical propagation of the two photons through separate optical fibers.

In realistic fiber-based quantum communication systems, the two photons propagate through distinct optical paths. As a consequence, polarization degradation may occur independently in the two propagation arms. This motivates the introduction of local depolarizing channels acting separately on subsystems A and B.

Effective local channel model. The effective evolution is modeled through independent local channels acting on the two subsystems:

$$\rho_{AB}^{\mathrm{out}} = (\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}) \left(\rho_{AB}^{\mathrm{in}}\right), \tag{5.3}$$

where $\mathcal{D}_{p_A}$ and $\mathcal{D}_{p_B}$ denote single-qubit depolarizing channels characterized by independent depolarization parameters p_A and p_B.

It is important to emphasize that the local channels act on the full bipartite density operator ρ_{AB}, rather than independently on the reduced states ρ_A and ρ_B.

Although the physical propagation occurs locally in each arm, the relevant quantum correlations are encoded in the nonlocal structure of the joint density operator. In particular, entanglement and bipartite correlation tensors cannot, in general, be reconstructed from the reduced density operators alone.

For maximally entangled Bell states, for example, the reduced states remain maximally mixed ($\rho_A = \rho_B = I/2$), independently of the entanglement structure of the global state. Consequently, evolving only the reduced density operators would fail to capture the degradation of bipartite quantum correlations induced by local propagation noise.

The tensor-product channel structure in Eq. (5.3) therefore preserves the full operator structure of the bipartite state while still representing physically local propagation effects.

For this reason, the action of a local channel on the bipartite state must be understood as an operator-map action on one tensor factor, not as an ordinary matrix multiplication. If a bipartite operator is written as a linear combination of product operators,

$$X_{AB} = \sum_k A_k \otimes B_k.$$

Using the single-qubit operator-map form of the depolarizing channel, a channel acting locally on subsystem A is applied as:

$$(\mathcal{D}_{p_A} \otimes \mathcal{I}) (X_{AB}) = \sum_k \mathcal{D}_{p_A}(A_k) \otimes B_k. \tag{5.4}$$

This representation is always possible in finite-dimensional bipartite systems, since product operators span the composite operator space.

Using the single-qubit operator-map form of the depolarizing channel,

$$\mathcal{D}_p(X) = (1-p)X + p\frac{I_2}{2}\operatorname{Tr}(X). \tag{5.5}$$

one obtains the single-arm action

$$(\mathcal{D}_{p_A} \otimes \mathcal{I}) (\rho_{AB}) = (1-p_A)\rho_{AB} + p_A\left(\frac{I_2}{2} \otimes \rho_B\right), \tag{5.6}$$

where $\rho_B = \operatorname{Tr}_A(\rho_{AB})$.

Analogously,

$$(\mathcal{I} \otimes \mathcal{D}_{p_B})(\rho_{AB}) = (1-p_B)\rho_{AB} + p_B\left(\rho_A \otimes \frac{I_2}{2}\right), \tag{5.7}$$

with $\rho_A = \mathrm{Tr}_B(\rho_{AB})$.

Applying the two single-arm channels in sequence gives

$$\begin{aligned}(\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B})(\rho_{AB}) = {} & (1-p_A)(1-p_B)\rho_{AB} \\ & + p_A(1-p_B)\left(\frac{I_2}{2} \otimes \rho_B\right) \\ & + (1-p_A)p_B\left(\rho_A \otimes \frac{I_2}{2}\right) \\ & + p_A p_B\left(\frac{I_2}{2} \otimes \frac{I_2}{2}\right).\end{aligned} \tag{5.8}$$

This expression makes explicit that local depolarization is local in its physical action, but it is still applied to the full joint state. The reduced density operators appear only because complete depolarization of one subsystem replaces that subsystem with $I_2/2$ while preserving the marginal state of the other subsystem.

This type of bipartite local channel has been studied extensively in the theory of composite quantum channels, including the characterization of positivity, complete positivity, entanglement-breaking, and entanglement-annihilating regimes for tensor products of depolarizing maps [6].

Single-qubit depolarizing action. For a single-qubit density operator represented in Bloch form as $\rho = \frac{1}{2}(I + \mathbf{r}\cdot\boldsymbol{\sigma})$, the depolarizing channel is modeled through

$$\mathcal{D}_p(\rho) = \frac{1}{2}(I + (1-p)\,\mathbf{r}\cdot\boldsymbol{\sigma}). \tag{5.9}$$

As discussed previously, the depolarizing channel acts as a linear operator map on the space of density operators rather than as an ordinary matrix multiplication. In the Bloch representation, this action corresponds to an isotropic contraction of the polarization vector.

Therefore, the depolarizing process cannot be interpreted simply as multiplication of the density operator by a scalar matrix factor. Instead, the channel replaces part of the operator structure by the maximally mixed contribution while preserving trace normalization.

This operator-map structure becomes particularly important in the bipartite setting, where local depolarization acts on the full joint density operator and modifies the correlation structure of the composite state.

The action of the channel therefore corresponds to an isotropic contraction of the Bloch vector:

$$\mathbf{r} \to (1-p)\mathbf{r}.$$

Consequently:

- $p = 0$ corresponds to ideal coherent propagation,
- $0 < p < 1$ describes partial polarization degradation,
- $p = 1$ maps the state to the maximally mixed state $I/2$.

This representation provides a simple and analytically tractable effective description of polarization decoherence while preserving a direct geometric interpretation at the level of the Bloch sphere.

Difference from global depolarization. It is important to distinguish local two-arm depolarization from the global depolarizing model introduced previously.

In the global model, the depolarizing map acts collectively on the full Hilbert space. For a d-dimensional system, the corresponding operator-map form is

$$\mathcal{D}_p^{(d)}(X) = (1-p)X + p\frac{I_d}{d}\,\mathrm{Tr}(X). \tag{5.10}$$

For a normalized density operator ρ, this reduces to

$$\mathcal{D}_p^{(d)}(\rho) = (1-p)\rho + p\frac{I_d}{d}. \tag{5.11}$$

In the two-qubit case considered here, $d = 4$, so that

$$\rho \to (1-p)\rho + \frac{p}{4}I_4. \tag{5.12}$$

Here the depolarization acts collectively on the full bipartite system.

In contrast, the local model acts independently on the two physical propagation arms through the tensor-product channel

$$\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}.$$

Although both models produce suppression of quantum correlations and progressive entanglement degradation, they correspond to physically distinct propagation mechanisms.

The local model is therefore more naturally aligned with the physical structure of fiber-based bipartite propagation, where each photon experiences an independent effective environment during transmission.

Role in the modeling hierarchy. The local depolarizing model constitutes the first effective fiber-channel model in which independent incoherent degradation mechanisms are explicitly associated with the two propagation arms.

This construction prepares the introduction of more advanced effective models combining coherent phase evolution and local depolarization within a unified composite propagation channel.

5.3 Correlation-Tensor Contraction Under Local Depolarization

The local depolarizing model introduced in the previous sections has a direct and experimentally meaningful effect on the bipartite correlation structure of the propagated state. Since polarization correlations are represented through expectation values of Pauli-product observables, the correlation tensor provides the natural framework for describing how local fiber degradation modifies measurable quantities.

The effective two-arm depolarizing channel is

$$\mathcal{E}_{p_A,p_B} = \mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}, \tag{5.13}$$

where $\mathcal{D}_{p_A}$ and $\mathcal{D}_{p_B}$ act independently on subsystems A and B.

The single-qubit depolarizing channel preserves the identity operator and contracts all traceless Pauli components isotropically:

$$\mathcal{D}_p(I) = I, \qquad \mathcal{D}_p(\sigma_i) = (1-p)\sigma_i, \qquad i \in \{x, y, z\}. \tag{5.14}$$

Consequently, for a generic two-body Pauli product $\sigma_i \otimes \sigma_j$, one obtains

$$\begin{aligned}\mathcal{E}_{p_A,p_B}(\sigma_i \otimes \sigma_j) &= \mathcal{D}_{p_A}(\sigma_i) \otimes \mathcal{D}_{p_B}(\sigma_j) \\ &= (1-p_A)(1-p_B)\sigma_i \otimes \sigma_j.\end{aligned} \tag{5.15}$$

It follows that all bipartite Pauli correlations are contracted by the common factor

$$\eta = (1-p_A)(1-p_B). \tag{5.16}$$

For a generic two-qubit state, the correlation tensor components are defined by

$$T_{ij} = \mathrm{Tr}\left(\rho_{AB}\sigma_i \otimes \sigma_j\right), \qquad i, j \in \{x, y, z\}. \tag{5.17}$$

Using Eq. (5.15), the output tensor satisfies

$$T_{ij}^{\mathrm{out}} = \eta T_{ij}^{\mathrm{in}}, \qquad i, j \in \{x, y, z\}. \tag{5.18}$$

Equivalently, in matrix form,

$$T_{\mathrm{out}} = \eta T_{\mathrm{in}}. \tag{5.19}$$

The quantity η therefore represents the effective survival factor of bipartite polarization correlations under independent local depolarization in the two propagation arms.

Physical interpretation. The tensor-contraction law derived above provides a simple geometric interpretation of isotropic local depolarization. The local channels do not alter the structural form of the correlation tensor; instead, they uniformly reduce its magnitude.

Physically, this means that local depolarization suppresses the strength of polarization correlations without introducing preferred directions in polarization space. Within the isotropic approximation adopted here, all two-body correlations are degraded uniformly.

This behavior differs conceptually from coherent phase evolution. A local phase shift modifies the orientation of the correlation structure relative to a fixed measurement basis, while isotropic depolarization reduces the overall correlation amplitude.

Operational consequences. For arbitrary local measurement directions $\mathbf{a}$ and $\mathbf{b}$, the correlation function is

$$E(\mathbf{a},\mathbf{b}) = \mathbf{a}^T T \mathbf{b}.$$

Using Eq. (5.19), the corresponding output correlation function becomes

$$E_{\text{out}}(\mathbf{a},\mathbf{b}) = \eta E_{\text{in}}(\mathbf{a},\mathbf{b}). \tag{5.20}$$

Therefore, within the isotropic local-depolarization model, every two-arm polarization correlation is suppressed by the same multiplicative factor η.

Since Bell-inequality parameters such as the CHSH quantity are constructed from linear combinations of these correlation functions, the same contraction factor directly affects the corresponding nonlocality indicators.

The tensor-contraction law established in this section constitutes the key analytical ingredient for the composite phase–depolarization fiber model developed in the following section.

5.4 Composite Effective Fiber Channel

The models introduced in the previous sections describe two complementary physical mechanisms affecting polarization-entangled photon propagation in optical fibers.

On one hand, coherent phase evolution modifies the orientation of the bipartite correlation structure through local unitary transformations. On the other hand, local depolarizing channels reduce the magnitude of polarization correlations through effective non-unitary dynamics.

A realistic reduced fiber model should therefore combine both effects simultaneously. This motivates the introduction of a composite effective channel in which coherent phase evolution and local depolarization act together on the bipartite polarization state.

5.4.1 Definition of the Composite Channel

The effective evolution is modeled through the composition

$$\rho_{AB}^{\text{out}} = (\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}) \left[U_\theta \rho_{AB}^{\text{in}} U_\theta^\dagger\right], \tag{5.21}$$

where

$$U_\theta = U_A(\theta) \otimes U_B,$$

with

$$U_A(\theta) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix},$$

and

$$U_B = I.$$

The parameter θ represents the residual coherent phase shift accumulated during propagation, while the depolarization parameters p_A and p_B quantify the effective incoherent polarization degradation occurring independently in the two fiber arms.

Using the operator-map representation of the local depolarizing channel introduced previously in (5.5), the composite channel admits the explicit form

$$\begin{aligned}\mathcal{E}_{\theta,p_A,p_B}\left(\rho_{AB}^{\text{in}}\right) = {} & (1-p_A)(1-p_B) U_\theta \rho_{AB}^{\text{in}} U_\theta^\dagger \\ & + p_A(1-p_B)\left(\frac{I_2}{2} \otimes \mathrm{Tr}_A\left[U_\theta \rho_{AB}^{\text{in}} U_\theta^\dagger\right]\right) \\ & + (1-p_A)p_B\left(\mathrm{Tr}_B\left[U_\theta \rho_{AB}^{\text{in}} U_\theta^\dagger\right] \otimes \frac{I_2}{2}\right) \\ & + p_A p_B \frac{I_4}{4}.\end{aligned} \tag{5.22}$$

The first contribution preserves the coherently evolved bipartite state, while the remaining terms progressively replace local subsystems with maximally mixed states through independent depolarization processes acting on the two propagation arms.

As demonstrated in Appendix C.2, isotropic local depolarization commutes with the reduced phase-evolution model considered here. Explicitly,

$$(\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}) \left[U_\theta \rho U_\theta^\dagger\right] = U_\theta \left[(\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B})(\rho)\right] U_\theta^\dagger. \tag{5.23}$$

Therefore, within the isotropic approximation adopted in this work, the ordering of the coherent and incoherent contributions is not relevant.

This commutation property is not expected to hold in general for arbitrary noise models. In particular, anisotropic polarization degradation, axis-dependent decoherence mechanisms, or dynamically varying local fiber transformations may break the commutation relation between coherent evolution and non-unitary dynamics.

In such cases, the ordering of successive channel contributions becomes physically relevant, and the propagation may require a segmented or infinitesimal description in which coherent and incoherent effects are applied sequentially along the fiber evolution.

The isotropic approximation adopted here therefore represents a simplified effective regime in which the analytical structure of the model remains tractable while preserving the essential interplay between coherent phase evolution and bipartite correlation degradation.

5.4.2 Correlation Tensor of the Composite Model

For the phase-evolved Bell state

$$|\psi(\theta)\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle + e^{i\theta}|10\rangle\right), \tag{5.24}$$

the correlation tensor obtained previously is

$$T(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{5.25}$$

Using the tensor-contraction law derived in Section 5.3, the output tensor under local two-arm depolarization becomes

$$T_{\text{out}}(\theta, p_A, p_B) = \eta T(\theta), \qquad \eta = (1-p_A)(1-p_B). \tag{5.26}$$

Explicitly,

$$T_{\text{out}}(\theta, p_A, p_B) = \eta \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{5.27}$$

Therefore,

$$T_{xx}^{\text{out}} = \eta\cos\theta, \tag{5.28}$$
$$T_{xy}^{\text{out}} = -\eta\sin\theta, \tag{5.29}$$
$$T_{yx}^{\text{out}} = \eta\sin\theta, \tag{5.30}$$
$$T_{yy}^{\text{out}} = \eta\cos\theta, \tag{5.31}$$
$$T_{zz}^{\text{out}} = -\eta. \tag{5.32}$$

All remaining tensor components vanish within the reduced phase model considered here.

Equation (5.27) separates clearly the coherent and incoherent contributions of the effective fiber channel. The phase parameter θ rotates the transverse xy-correlation structure, while the depolarization factor η contracts the magnitude of the full bipartite tensor uniformly.

Thus, coherent phase evolution changes the orientation of the polarization correlations, whereas local depolarization suppresses their amplitude.

5.4.3 State Metrics Under the Composite Channel

The composite channel introduced above allows the simultaneous study of coherent phase mismatch and genuine non-unitary correlation degradation through quantitative state metrics.

Purity. For the composite phase–depolarization model, the global purity becomes

$$\gamma_{AB} = \mathrm{Tr}\left(\rho_{AB}^2\right) = \frac{1+3\eta^2}{4}, \tag{5.33}$$

where

$$\eta = (1-p_A)(1-p_B).$$

The purity depends only on the depolarization strength and is independent of the phase parameter θ. This reflects the fact that coherent unitary evolution preserves the eigenvalue spectrum of the density operator, while depolarization introduces genuine mixedness.

For maximally entangled Bell-state inputs, the reduced density operators remain maximally mixed:

$$\rho_A = \rho_B = \frac{I}{2}, \tag{5.34}$$

so that

$$\gamma_A = \gamma_B = \frac{1}{2}. \tag{5.35}$$

Bell-state fidelity. The fidelity with respect to the reference Bell state $|\Psi^+\rangle$ becomes

$$F_{\Psi^+} = \frac{1-\eta}{4} + \eta\frac{1+\cos\theta}{2}, \tag{5.36}$$

The first contribution originates from the isotropic mixed component introduced by local depolarization, while the second term corresponds to the coherent overlap of the phase-evolved Bell state with the chosen Bell reference basis.

The phase parameter θ modifies the relative orientation of the state with respect to the reference Bell basis and may therefore reduce fidelity even in the absence of entanglement degradation. In contrast, the depolarization factor η reduces the strength of the bipartite quantum correlations themselves.

Consequently, reduced Bell-state fidelity does not necessarily imply reduced entanglement.

Concurrence. For the locally depolarized Bell-state family considered here, the concurrence is

$$C = \max\left[0, \frac{3\eta - 1}{2}\right]. \tag{5.37}$$

The concurrence is independent of the phase parameter θ, since local unitary transformations do not modify entanglement.

Therefore, coherent phase evolution changes the observed correlation structure while preserving bipartite entanglement, whereas local depolarization produces genuine entanglement degradation.

Entanglement disappears when

$$\eta = \frac{1}{3}. \tag{5.38}$$

For symmetric local depolarization,

$$p_A = p_B = p,$$

this condition becomes

$$(1 - p)^2 = \frac{1}{3}, \tag{5.39}$$

which gives

$$p = 1 - \frac{1}{\sqrt{3}} \approx 0.423. \tag{5.40}$$

CHSH nonlocality. Since arbitrary-axis correlation functions scale according to

$$E_{\mathrm{out}} = \eta E_{\mathrm{in}},$$

the maximal CHSH parameter also contracts linearly:

$$S_{\max}^{\mathrm{out}} = \eta S_{\max}^{\mathrm{in}}. \tag{5.41}$$

For a maximally entangled Bell state,

$$S_{\max}^{\mathrm{in}} = 2\sqrt{2},$$

so that

$$S_{\max}^{\mathrm{out}} = 2\sqrt{2}\,\eta. \tag{5.42}$$

Bell-inequality violation requires

$$S_{\max}^{\mathrm{out}} > 2, \tag{5.43}$$

which implies

$$\eta > \frac{1}{\sqrt{2}}. \tag{5.44}$$

For symmetric local depolarization,

$$p < 1 - 2^{-1/4} \approx 0.159. \tag{5.45}$$

Therefore, CHSH nonlocality disappears before entanglement vanishes completely, showing that Bell-inequality violation is more fragile than concurrence under local depolarization.

5.4.4 Physical Interpretation

The composite channel developed in this section provides the first unified effective fiber model introduced in this work.

Within this framework, the phase parameter θ represents residual coherent birefringent evolution, while the depolarization parameters p_A and p_B describe effective incoherent polarization degradation occurring independently in the two propagation arms.

The correlation tensor acts as the bridge between the abstract density-operator description and experimentally accessible polarization measurements. Coherent phase evolution rotates the tensor structure relative to a fixed laboratory basis, whereas local depolarization contracts the tensor magnitude and progressively suppresses bipartite correlations.

The resulting model combines coherent polarization rotation and genuine non-unitary correlation degradation within a single analytically tractable framework.

This composite description constitutes the theoretical foundation for the numerical investigations developed in the following chapter, where the analytical predictions derived above are validated through simulation.

5.5 Statistical Composite Fiber Channel

The composite fiber model introduced in the previous section describes the simultaneous action of coherent phase evolution and local isotropic depolarization for a fixed phase parameter θ.

In realistic propagation scenarios, however, the relative phase accumulated during transmission may fluctuate statistically between different realizations. Consequently, the effective propagated state must combine both statistical phase averaging and local depolarizing degradation within a unified reduced description.

The purpose of this section is to extend the composite coherent-phase channel toward a statistical effective model capable of describing simultaneous coherent uncertainty and incoherent polarization degradation.

5.5.1 Ensemble-Averaged Composite Channel

Consider a statistical ensemble of phase realizations θ_k with associated probabilities p_k. The corresponding ensemble-averaged propagated state is

$$\rho_{AB}^{(\mathrm{ens})} = \sum_k p_k \left(\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}\right) \left[U_{\theta_k} \rho_{AB}^{(\mathrm{in})} U_{\theta_k}^{\dagger}\right]. \tag{5.46}$$

Using the commutation property established previously, the local depolarizing channel may be moved outside the statistical average:

$$\rho_{AB}^{(\mathrm{ens})} = \left(\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}\right) \left[\sum_k p_k U_{\theta_k} \rho_{AB}^{(\mathrm{in})} U_{\theta_k}^{\dagger}\right]. \tag{5.47}$$

Therefore, the statistical phase averaging and the isotropic local depolarization remain analytically separable within the effective model adopted in this work.

Following the notation introduced previously, define the phase-coherence parameter

$$\mu = \left\langle e^{i\theta} \right\rangle = \sum_k p_k e^{i\theta_k}. \tag{5.48}$$

Its real and imaginary parts satisfy

$$\mathrm{Re}(\mu) = \langle \cos\theta \rangle, \qquad \mathrm{Im}(\mu) = \langle \sin\theta \rangle. \tag{5.49}$$

The quantity $|\mu|$ therefore measures the residual phase coherence surviving the ensemble averaging process.

5.5.2 Correlation Tensor of the Statistical Composite Model

For the random-phase ensemble considered previously, the phase averaging modifies only the transverse correlation structure. Including local isotropic depolarization yields the effective tensor

$$T_{\mathrm{out}}^{(\mathrm{ens})} = \eta \begin{pmatrix} \mathrm{Re}(\mu) & -\mathrm{Im}(\mu) & 0 \\ \mathrm{Im}(\mu) & \mathrm{Re}(\mu) & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{5.50}$$

Explicitly,

$$T_{xx}^{(\mathrm{ens})} = \eta\, \mathrm{Re}(\mu), \tag{5.51}$$
$$T_{xy}^{(\mathrm{ens})} = -\eta\, \mathrm{Im}(\mu), \tag{5.52}$$
$$T_{yx}^{(\mathrm{ens})} = \eta\, \mathrm{Im}(\mu), \tag{5.53}$$
$$T_{yy}^{(\mathrm{ens})} = \eta\, \mathrm{Re}(\mu), \tag{5.54}$$
$$T_{zz}^{(\mathrm{ens})} = -\eta. \tag{5.55}$$

All remaining tensor components vanish.

Equation (5.50) separates clearly the effects of the two physical mechanisms entering the model.

The statistical phase distribution modifies the transverse coherence structure through the parameter μ, while the local depolarizing channel contracts all bipartite correlations uniformly through the factor η.

An important consequence is that the longitudinal anticorrelation term T_{zz} is unaffected by phase averaging itself and is reduced only through local depolarization.

Therefore, phase fluctuations and isotropic depolarization leave distinct signatures on the correlation tensor structure.

5.5.3 State Metrics of the Statistical Composite Model

The statistical composite channel allows coherent uncertainty and incoherent polarization degradation to contribute simultaneously to the observable state metrics.

Purity. For the random-phase ensemble without local depolarization, the purity satisfies

$$\gamma\left(\rho_{AB}^{(\mathrm{ens})}\right) = \frac{1+|\mu|^2}{2}.$$

Including local isotropic depolarization yields

$$\gamma\left(\rho_{AB}^{(\mathrm{ens})}\right) = \frac{1+\eta^2\left(1+2|\mu|^2\right)}{4}. \tag{5.56}$$

The purity therefore decreases both when the phase coherence parameter $|\mu|$ is reduced and when the depolarization factor η decreases.

Bell-state fidelity. The Bell-state fidelity with respect to $|\Psi^+\rangle$ becomes

$$F_{\Psi^+} = \frac{1-\eta}{4} + \eta\frac{1+\mathrm{Re}(\mu)}{2}, \tag{5.57}$$

The first contribution originates from the isotropic mixed component introduced by local depolarization, while the second term represents the coherent overlap associated with the residual ensemble phase coherence.

Therefore, the fidelity depends simultaneously on $\mathrm{Re}(\mu)$, which quantifies residual phase alignment with the reference Bell basis, and on the depolarization factor η, which globally contracts the bipartite correlations.

Consequently, reduced Bell-state fidelity may originate either from statistical phase fluctuations, from incoherent depolarization, or from both simultaneously.

Concurrence. For the statistical composite model, the concurrence becomes

$$C = \max\left[0, \eta|\mu| - \frac{1-\eta}{2}\right]. \tag{5.58}$$

Therefore, both statistical phase averaging and local depolarization contribute to entanglement degradation.

Phase averaging suppresses the coherence amplitude through $|\mu|$, while local depolarization contracts the bipartite correlations globally through the factor η.

CHSH nonlocality. Since the correlation tensor is globally contracted by η, the maximal CHSH parameter scales according to

$$S_{\max}^{(\mathrm{ens})} = 2\eta\sqrt{1+|\mu|^2}. \tag{5.59}$$

Bell-inequality violation therefore requires

$$\eta\sqrt{1+|\mu|^2} > 1. \tag{5.60}$$

Consequently, both reduced phase coherence and local depolarization suppress non-local quantum correlations.

5.5.4 Physical Interpretation

The statistical composite channel developed in this section provides the most general effective fiber model considered in the present work.

Within this framework, two physically distinct mechanisms contribute simultaneously to the degradation of bipartite polarization correlations.

The statistical phase distribution suppresses transverse coherence through ensemble averaging, reducing the effective phase-coherence parameter $|\mu|$. In contrast, isotropic local depolarization contracts the full correlation tensor uniformly through the factor η.

These mechanisms therefore affect the correlation structure differently. Phase averaging primarily modifies the coherent transverse interference structure, whereas local depolarization reduces the global magnitude of bipartite polarization correlations.

The resulting tensor structure provides a direct connection between microscopic propagation fluctuations and experimentally observable polarization-correlation measurements.

Most importantly, the model shows that coherent uncertainty and incoherent polarization degradation leave distinguishable signatures on entanglement, Bell-state fidelity, correlation tensors, and CHSH nonlocality.

This statistical composite framework constitutes the theoretical basis for the final numerical investigations developed in chapter 6.

Chapter 6

Numerical Experiments and Results

This chapter presents the numerical validation of the theoretical models developed in the previous chapters. The experiments are organized according to a progressive hierarchy of physical complexity, starting from fixed unitary phase evolution and then extending toward stochastic and non-unitary propagation effects.
The objective is twofold: first, to verify the consistency between analytical predictions and numerical implementation; second, to investigate how different propagation mechanisms affect observable correlations, state purity, fidelity, concurrence, and nonlocality.
The numerical study follows the modeling hierarchy:

$$\begin{array}{c}\text{Fixed Phase Evolution}\\ \downarrow \\ \text{Statistical Phase Fluctuations}\\ \downarrow \\ \text{Effective Depolarization}\\ \downarrow \\ \text{CHSH Nonlocality}\end{array}$$

6.1 Experiment 1: Fixed Phase Sweep

Objective and Physical Motivation

This first experiment establishes the baseline propagation regime in which the optical fiber acts as an effective phase shifter preserving both the purity and the entanglement of the bipartite state.
The objective is to characterize how observable correlations and state-characterization quantities evolve under the reduced unitary fiber model introduced in Section 3.5.
This experiment represents the minimal physically relevant propagation scenario. The fiber is modeled exclusively through a fixed differential phase shift acting on the polarization components of the transmitted photons, while all non-unitary effects are neglected.
Consequently, the experiment isolates the role of coherent phase evolution and establishes the baseline regime against which later noisy and depolarizing models will be compared.

Model and Analytical Predictions

The initial bipartite state is chosen as the Bell state

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle + |10\rangle\right).$$

Under the reduced phase model, the fiber induces an effective relative phase shift between the polarization components, leading to the evolved state

$$|\psi(\theta)\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle + e^{i\theta}|10\rangle\right). \tag{6.1}$$

The diagonal correlation observables considered in the simulation are

$$c_{xx} = \langle \sigma_x \otimes \sigma_x \rangle, \qquad c_{yy} = \langle \sigma_y \otimes \sigma_y \rangle, \qquad c_{zz} = \langle \sigma_z \otimes \sigma_z \rangle.$$

Direct analytical evaluation gives

$$c_{xx} = \cos\theta, \qquad c_{yy} = \cos\theta, \qquad c_{zz} = -1.$$

Since the evolution is generated by local unitary transformations, the global state remains pure:

$$\gamma(\rho(\theta)) = \mathrm{Tr}\left(\rho(\theta)^2\right) = 1.$$

The reduced density operators remain maximally mixed:

$$\rho_A(\theta) = \rho_B(\theta) = \frac{I}{2},$$

which implies

$$\gamma(\rho_A) = \gamma(\rho_B) = \frac{1}{2}.$$

The fidelity with respect to the reference Bell state $|\Psi^+\rangle$ is

$$F_{\Psi^+}(\rho(\theta)) = \frac{1+\cos\theta}{2}.$$

The concurrence remains constant:

$$C(\rho(\theta)) = 1,$$

demonstrating the invariance of bipartite entanglement under local unitary evolution.

The experiment therefore provides a controlled setting in which observable correlations evolve continuously with the phase parameter θ, while the underlying entanglement structure remains unchanged.

Numerical Results

A numerical sweep over the interval

$$\theta \in [0, 2\pi]$$

is performed in order to evaluate both correlation observables and state-characterization quantities throughout the phase evolution.
Figure 6.1 compares the numerical results obtained from the simulator with the corresponding analytical predictions for the diagonal correlation observables.

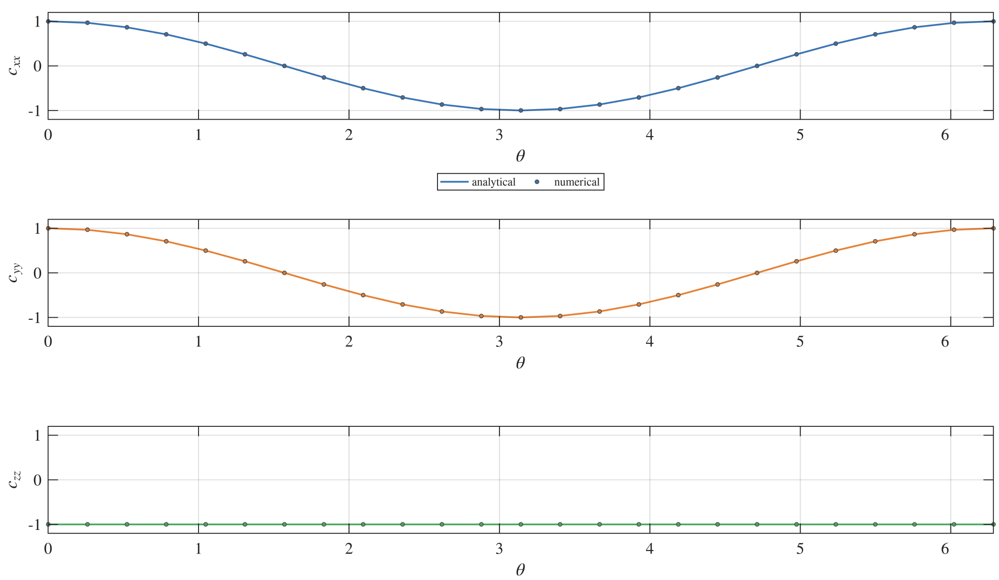


Figure 6.1: Numerical and analytical comparison for the correlation observables c_{xx}, c_{yy}, and c_{zz}. The agreement validates the implementation of the reduced phase model.

The numerical curves reproduce the expected cosine dependence for c_{xx} and c_{yy}, while c_{zz} remains constant at -1 over the entire propagation range.
The numerical and analytical curves are visually indistinguishable within numerical precision.

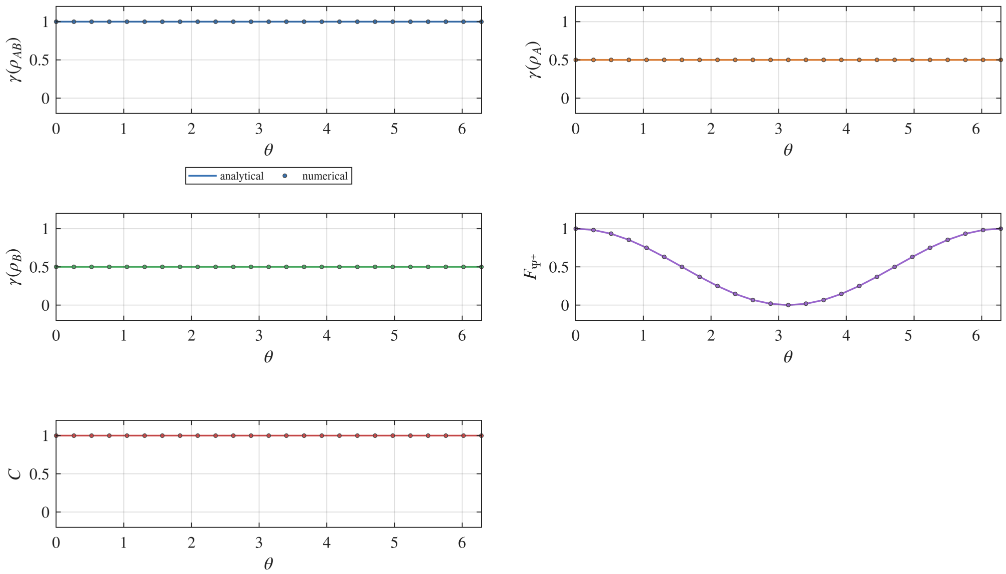


Figure 6.2: State-characterization quantities for the phase-evolved state. Global purity and concurrence remain constant, while fidelity varies with the relative phase θ.

Figure 6.2 shows the evolution of the state-characterization quantities.
The global purity remains equal to unity throughout the evolution, confirming that the propagated state remains pure. The reduced purities remain fixed at 1/2, while the concurrence remains equal to 1, indicating preservation of maximal entanglement.
In contrast, the fidelity with respect to the reference Bell state varies continuously with the phase parameter.

The full correlation tensor obtained numerically is shown in Figure 6.3.
The numerical results confirm the analytical tensor structure

$$T(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{6.2}$$

The non-zero off-diagonal components

$$T_{xy} = -\sin\theta, \qquad T_{yx} = \sin\theta$$

demonstrate that the phase evolution induces a rotation of the correlation structure within the transverse x-y plane.
This structure is not visible when considering only diagonal observables, but emerges naturally within the full tensor representation.

Discussion

The numerical results are in full agreement with the analytical predictions.
The experiment confirms that phase evolution modifies measurable two-photon correlations while preserving both the purity and the entanglement of the global state.

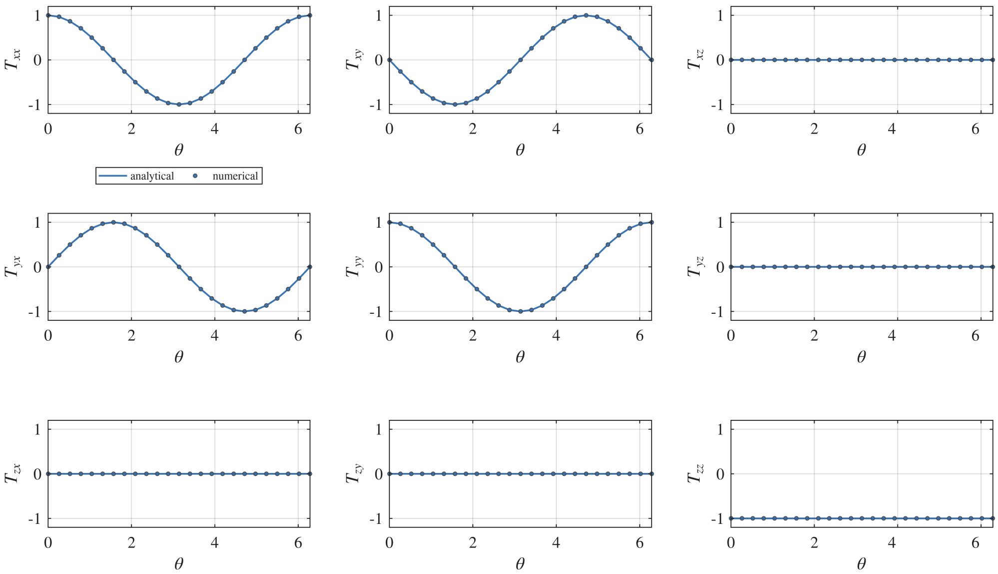


Figure 6.3: Full correlation tensor $T_{ij}(\theta)$ for the phase-evolved state. The off-diagonal tensor components reveal the rotation of the correlation structure in the transverse plane.

In particular, the transverse correlations c_{xx} and c_{yy} exhibit the expected cosine dependence on the phase parameter θ, whereas c_{zz} remains invariant.
At the same time, the concurrence remains constant despite the continuous variation of the fidelity with respect to the reference Bell state. This distinction is physically significant.
The fidelity quantifies the overlap with a specific reference state and therefore depends explicitly on the relative phase. The concurrence, instead, measures the intrinsic entanglement content of the bipartite state and remains invariant under local unitary transformations.
Consequently, changes in fidelity do not necessarily imply degradation of entanglement.
The experiment therefore establishes a clear separation between:

- modifications of observable correlations induced by coherent phase evolution;
- genuine loss of entanglement caused by non-unitary processes.

This distinction will become essential in the interpretation of the later experiments involving statistical averaging and depolarization.

Physical Interpretation

The reduced phase model admits a direct physical interpretation in terms of fiber-induced birefringence.
In optical fibers, the horizontal and vertical polarization components generally propagate with different effective refractive indices. As a consequence, the two polarization modes accumulate different propagation phases.

After compensation of global polarization rotations, the dominant residual effect is a differential phase shift between the polarization components. Within the qubit representation, this residual evolution corresponds to a local unitary rotation around the Z-axis of the Bloch sphere.
The phase parameter θ therefore represents the effective relative phase accumulated during propagation.
The experiment shows that this mechanism modifies measurable correlations without degrading the entanglement of the transmitted state. The global quantum state remains pure, and the bipartite entanglement remains maximal throughout the evolution.
At the same time, the dependence of the transverse correlations on θ demonstrates that experimentally accessible observables remain highly sensitive to phase fluctuations.
This experiment therefore establishes the minimal effective model of coherent fiber propagation: a deterministic unitary transformation that continuously rotates the observable correlation structure while preserving the underlying entangled nature of the bipartite quantum state.

6.2 Experiment 2: Random Phase Ensemble Sweep

Objective and Physical Motivation

The second experiment extends the fixed-phase model by introducing random phase realizations θ_k, modeling stochastic birefringence effects in realistic optical fibers. Unlike the coherent regime studied in Section 6.1, the system is no longer described by a single pure state, but by a statistical ensemble of phase-evolved realizations

$$\{|\psi(\theta_k)\rangle\}_{k=1}^{N},$$

together with the corresponding ensemble-averaged density operator.
The objective is to characterize how unresolved phase fluctuations modify observable correlations, global coherence, purity, fidelity, and entanglement.
This experiment therefore represents the transition from purely deterministic unitary evolution toward an effective mixed-state description arising from statistical averaging.

Model and Analytical Predictions

Each phase realization is described by the pure state

$$|\psi(\theta_k)\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle + e^{i\theta_k}|10\rangle\right). \tag{6.3}$$

The ensemble-averaged state is then constructed as

$$\rho^{(\mathrm{ens})} = \frac{1}{N}\sum_{k=1}^{N} |\psi(\theta_k)\rangle\langle\psi(\theta_k)|. \tag{6.4}$$

Introducing the ensemble coherence parameter

$$\mu = \langle e^{i\theta}\rangle = \langle\cos\theta\rangle + i\langle\sin\theta\rangle, \tag{6.5}$$

the ensemble-averaged diagonal correlations become

$$c_{xx} = \mathrm{Re}(\mu), \qquad c_{yy} = \mathrm{Re}(\mu), \qquad c_{zz} = -1.$$

The global purity is

$$\gamma(\rho^{\mathrm{ens}}) = \frac{1+|\mu|^2}{2}.$$

The reduced density operators remain maximally mixed:

$$\rho_A = \rho_B = \frac{I}{2},$$

which implies

$$\gamma(\rho_A) = \gamma(\rho_B) = \frac{1}{2}.$$

The fidelity with respect to the reference Bell state and the concurrence are given by

$$F_{\Psi^+} = \frac{1 + \mathrm{Re}(\mu)}{2}, \qquad C(\rho^{(\mathrm{ens})}) = |\mu|.$$

The ensemble coherence parameter μ therefore provides a compact description of the entire propagation regime. Its real part determines observable transverse correlations and fidelity, while its magnitude controls the residual entanglement of the averaged state.

Numerical Results

Three representative phase distributions are considered:

- constant phase distribution;
- uniform phase distribution;
- Gaussian phase distribution.

For each case, the simulator evaluates:

- sampled phase statistics;
- pure-state correlation trends;
- ensemble-averaged correlations;
- purity, fidelity, and concurrence.

Figure 6.4 shows the constant-phase regime.

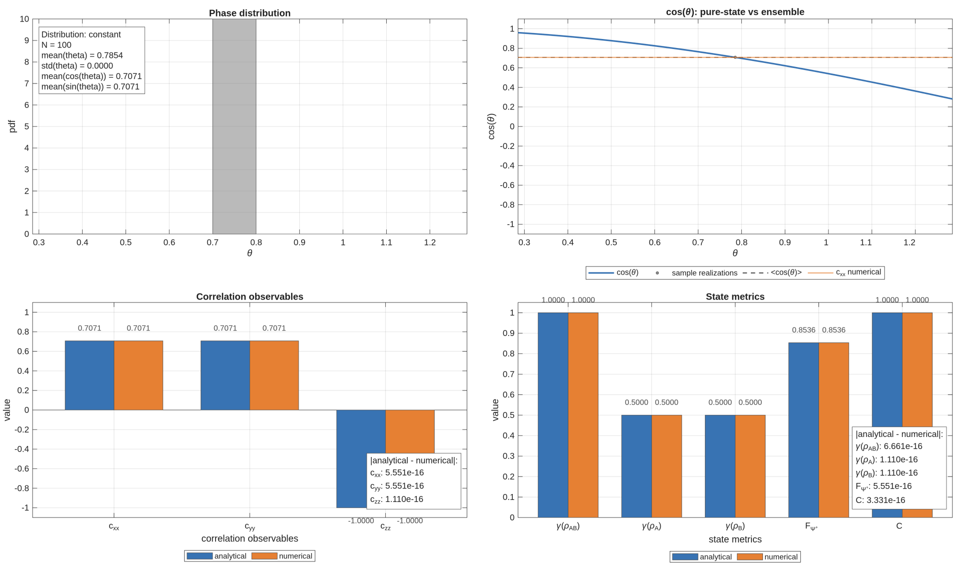


Figure 6.4: Constant phase distribution: the ensemble reduces to a coherent realization and concurrence remains maximal.

In this limit, all realizations coincide and the ensemble reproduces the coherent behavior observed in Experiment 1.

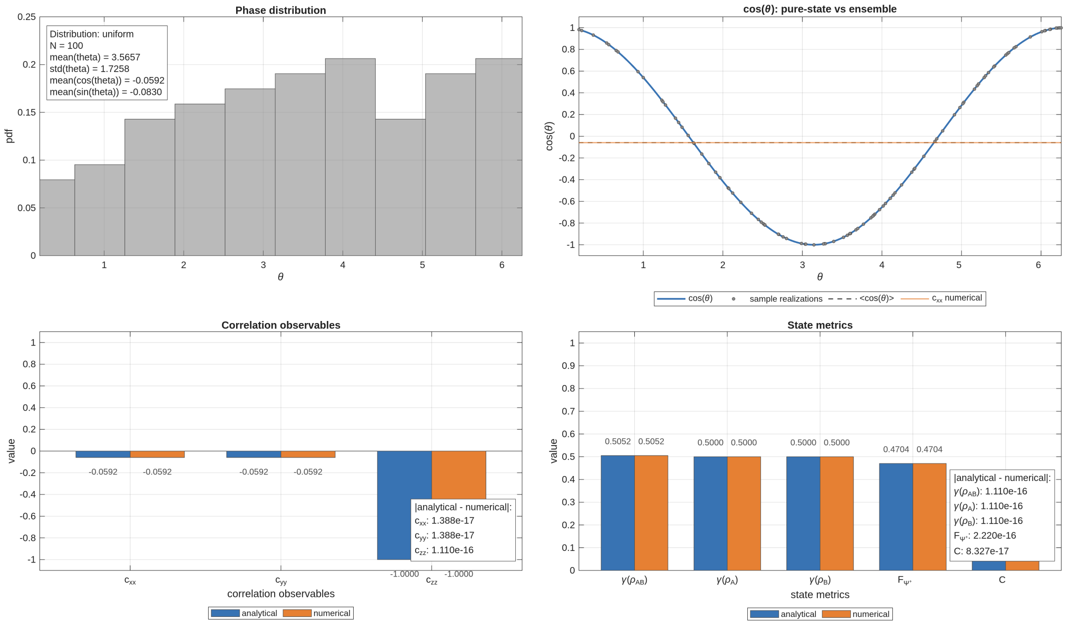


Figure 6.5: Uniform phase distribution: transverse correlations vanish, fidelity decreases, and concurrence tends to zero.

Figure 6.5 shows the behavior corresponding to a uniform phase distribution.
The averaging process suppresses the transverse coherence terms, leading to vanishing ensemble correlations in the x–y plane and to a strong reduction of both fidelity and concurrence.

Figure 6.6 presents the intermediate regime generated by a Gaussian phase distribution.
In this case, residual coherence survives ensemble averaging, producing intermediate values of purity, fidelity, and concurrence.

The corresponding correlation tensor structures are shown in Figure 6.7.
As the number of realizations increases, the numerically reconstructed tensor components converge toward their analytical expectation values.
At the same time, tensor entries that are analytically zero remain numerically negligible throughout the simulation, confirming the internal consistency of the ensemble implementation.

Discussion

The numerical results reveal a progressive transition from coherent evolution to ensemble-induced decoherence.
The constant-phase regime reproduces the behavior observed in Experiment 1, where the ensemble effectively reduces to a single pure realization.
As the phase distribution broadens, the transverse coherence terms are progressively suppressed. Consequently, the ensemble coherence parameter satisfies

$$|\mu| \to 0,$$

leading to the disappearance of coherent transverse correlations.
This transition is reflected directly in the state-characterization quantities:

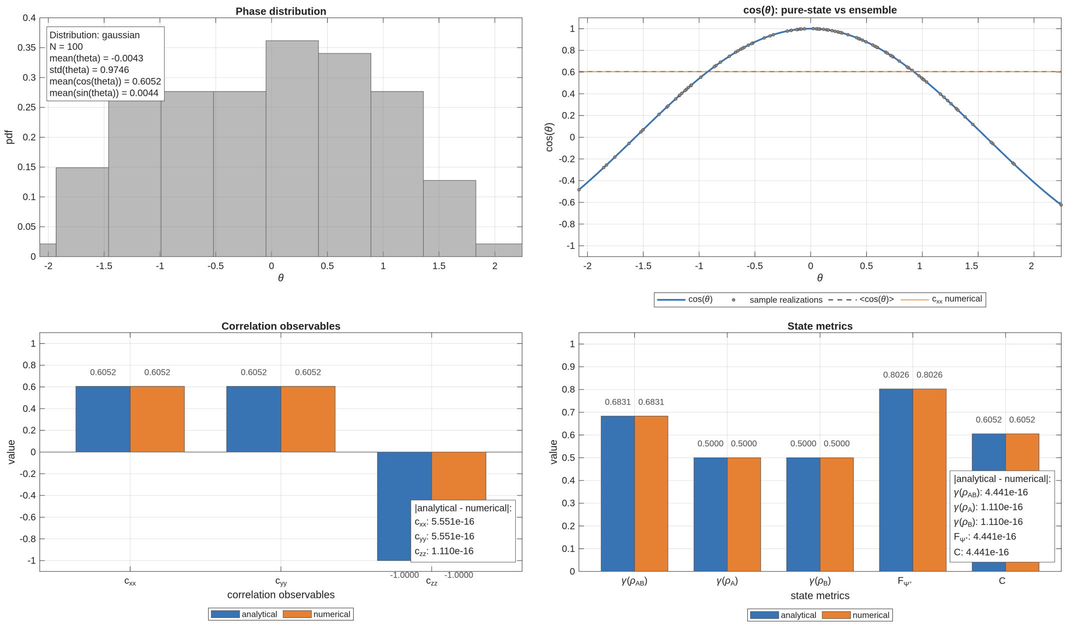
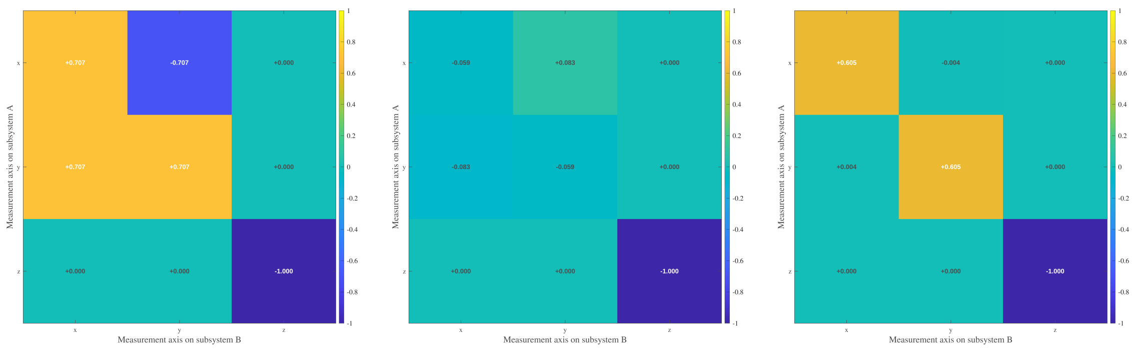


Figure 6.6: Gaussian phase distribution: partial coherence leads to intermediate values of purity and concurrence.



(a) Constant phase (b) Uniform distribution (c) Gaussian distribution

Figure 6.7: Correlation tensor snapshots for different phase distributions. Random phase averaging suppresses transverse coherence, progressively reducing the off-diagonal tensor components, while the longitudinal correlation $T_{zz} = -1$ remains invariant.

- the global purity decreases from unity;
- the reduced purities remain fixed at $1/2$;
- the fidelity decreases;
- the concurrence decreases continuously from 1 toward 0.

Unlike the fixed-phase evolution studied previously, the reduction of fidelity is now associated with a genuine degradation of entanglement.
The experiment therefore establishes an important conceptual distinction:

- fixed-phase evolution modifies observable correlations without destroying entanglement;
- unresolved statistical phase fluctuations generate effective mixedness and progressively suppress entanglement.

This mechanism constitutes the first non-unitary effect emerging naturally from the propagation model.

Physical Interpretation

Each individual realization of the ensemble remains a pure maximally entangled state. The loss of coherence arises only after averaging over unresolved phase fluctuations.
Physically, this situation models realistic fiber propagation conditions in which the accumulated birefringence-induced phase is not perfectly controlled and varies over time, frequency, or environmental conditions.
The ensemble coherence parameter

$$\mu = \langle e^{i\theta} \rangle$$

provides a unified physical interpretation of the propagation regime.
Its real part determines the observable transverse correlations and the fidelity with respect to the reference Bell state, while its magnitude quantifies the residual entanglement preserved after averaging.
As the phase fluctuations become broader, the ensemble loses its coherent phase structure. The corresponding density operator progressively approaches a mixed state, leading to a reduction of both purity and concurrence.
This experiment therefore captures the transition from ideal coherent propagation toward realistic noisy optical-fiber evolution, where stochastic birefringence effects induce effective decoherence and entanglement degradation.

6.3 Experiment 3: Global Depolarizing Channel

Objective and Physical Motivation

This experiment investigates the degradation of a maximally entangled Bell state under an effective two-qubit depolarizing channel.
Unlike the previous experiments, which were based on coherent unitary evolution or statistical averaging over unitary realizations, the present model introduces a genuinely non-unitary transformation acting directly at the level of the density operator.
The objective is to characterize how depolarization affects:

- purity and mixedness;
- fidelity with respect to the Bell reference state;
- concurrence and entanglement degradation;
- observable correlations;
- the structure of the correlation tensor.

This experiment therefore establishes the first explicit decoherence model of the simulator and provides a baseline description of irreversible coherence loss in bipartite quantum propagation.

Model and Analytical Predictions

The initial state is the Bell state

$$|\Psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}, \qquad \rho_0 = |\Psi^+\rangle\langle\Psi^+|.$$

The depolarizing channel is defined as

$$\rho(p) = (1-p)\rho_0 + \frac{p}{4}I_4, \qquad p \in [0,1]. \tag{6.6}$$

The parameter p controls the strength of the depolarization process.
The limiting cases are

$$\rho(0) = \rho_0, \qquad \rho(1) = \frac{I_4}{4}.$$

Thus, the channel continuously interpolates between a pure maximally entangled Bell state and the maximally mixed two-qubit state.

For this family of states, the analytical predictions are

$$\gamma_{AB}(p) = 1 - \frac{3}{2}p + \frac{3}{4}p^2, \qquad \gamma_A = \gamma_B = \frac{1}{2},$$

$$F_{\Psi^+}(p) = 1 - \frac{3}{4}p, \qquad C(p) = \max\left(0, 1 - \frac{3}{2}p\right).$$

The concurrence vanishes at

$$p = \frac{2}{3},$$

marking the transition from entangled to separable states.

The action of the depolarizing channel on the correlation tensor corresponds to a uniform contraction of the Bell-state tensor:

$$T(p) = (1-p)T_{\Psi^+}.$$

Using the Bell-state tensor

$$T_{\Psi^+} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix},$$

one obtains

$$T(p) = \begin{pmatrix} 1-p & 0 & 0 \\ 0 & 1-p & 0 \\ 0 & 0 & -(1-p) \end{pmatrix}. \tag{6.7}$$

The corresponding diagonal correlations are therefore

$$c_{xx}(p) = 1-p, \qquad c_{yy}(p) = 1-p, \qquad c_{zz}(p) = -(1-p).$$

These expressions are consistent with the correlation-tensor formalism introduced in Section 3.6 and with the analytical derivations reported in Appendix **??**.

Numerical Results

The depolarization parameter p is sampled over the interval

$$p \in [0,1],$$

and all state-characterization quantities and correlation observables are evaluated numerically and compared with the corresponding analytical predictions.
The numerical results show agreement at machine precision for all quantities considered in the simulation.
In particular:

- the Frobenius-norm error of the correlation tensor remains of order 10^{-16};
- the errors on purity, fidelity, concurrence, and diagonal correlations remain at numerical roundoff level.

Figure 6.8 summarizes the numerical evolution of the principal observables under the depolarizing channel.
The numerical and analytical curves are visually indistinguishable within numerical precision.
The global purity decreases monotonically from unity toward the maximally mixed limit, while the concurrence exhibits threshold behavior and vanishes at

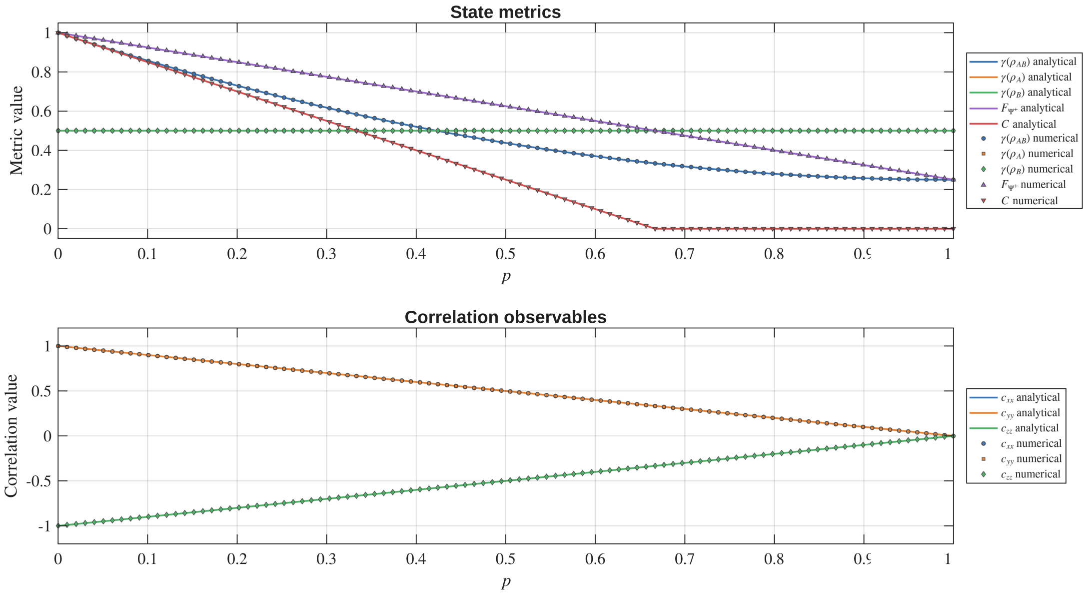


Figure 6.8: Numerical and analytical behavior of purity, fidelity, concurrence, and correlations under the depolarizing channel. Agreement is exact up to numerical precision.

$$p = \frac{2}{3}.$$

At the same time, the diagonal correlations decay linearly in magnitude, consistently with the tensor contraction law given in Eq. (6.7).

Discussion

This experiment validates the first explicit non-unitary extension of the simulator. Unlike the fixed-phase and ensemble-based phase models studied previously, the depolarizing channel introduces an isotropic loss of coherence directly at the level of the two-qubit density operator.
The numerical results show that the different observables exhibit distinct degradation behaviors:

- the global purity decreases smoothly over the full interval;
- the concurrence exhibits a sharp entanglement threshold;
- the observable correlations decay linearly;
- the full correlation tensor undergoes a uniform isotropic contraction.

The reduced density operators remain maximally mixed throughout the evolution:

$$\rho_A = \rho_B = \frac{I}{2}.$$

This behavior is consistent with the results of Experiment 6.2, where local mixedness also remained constant despite degradation of global coherence.

However, the physical origin of mixedness is fundamentally different in the two cases. In the random-phase ensemble model, mixedness arises from averaging over unresolved coherent realizations while preserving an underlying phase structure.
In contrast, the depolarizing channel suppresses the nontrivial two-qubit correlations isotropically by mixing the state directly with the maximally mixed operator $I_4/4$. The experiment therefore establishes a clear distinction between:

- decoherence emerging from statistical phase uncertainty;
- irreversible isotropic depolarization modeled through a CPTP channel.

Physical Interpretation

The depolarizing channel considered in this experiment is a global two-qubit depolarizing channel. It acts directly on the complete bipartite density operator and models the progressive replacement of the initial Bell state by the maximally mixed two-qubit state,
In this sense, the parameter p does not represent the degradation of a specific fiber arm. Instead, it controls an isotropic loss of information at the level of the full two-photon state. The limiting cases are directly interpretable: for $p = 0$, the initial Bell state is recovered, while for $p = 1$, the output state is completely mixed.

Physically, this model should be interpreted as an effective baseline for incoherent two-qubit degradation rather than as a microscopic description of optical-fiber propagation. It captures the idealized situation in which all nontrivial information contained in the bipartite state is uniformly suppressed, without distinguishing between different physical origins or between the two propagation arms.

Under global depolarization, all traceless components of the two-qubit state are contracted uniformly. As a consequence, the observable two-photon correlations, the fidelity with respect to the reference Bell state, and the concurrence decrease monotonically as p increases. In the fully depolarized limit, the state $I_4/4$ contains no preferred polarization structure, no entanglement, and no nonzero two-qubit correlation tensor components.

This behavior differs from the random-phase ensemble studied previously. In the random-phase model, decoherence arises from statistical averaging over unresolved phase realizations and therefore preserves a specific phase-related structure. By contrast, the global depolarizing channel removes information isotropically: no particular phase direction, measurement basis, or fiber arm is privileged.

For this reason, the global depolarizing channel is not intended to be the final physical model of fiber propagation. Its role is to provide a simple and analytically transparent reference case for mixed-state degradation. More realistic descriptions, such as local depolarization acting independently on arms A and B, will refine this baseline by assigning the loss mechanism to the individual propagation paths.

6.4 Experiment 4: CHSH Nonlocality

Objective and Physical Motivation

This experiment investigates the nonlocal correlations of the bipartite quantum state through the Clauser–Horne–Shimony–Holt (CHSH) parameter.
The analysis compares two physically distinct propagation mechanisms:

- fixed-phase evolution generated by local unitary transformations;
- depolarizing noise described by an effective two-qubit quantum channel.

The objective is to determine how these mechanisms affect the observable violation of the CHSH inequality and to distinguish between:

- apparent reductions of CHSH violation caused by measurement-axis misalignment;
- genuine loss of nonlocality caused by decoherence.

To this end, two different CHSH quantities are evaluated:

- S_{fixed}, computed using fixed laboratory measurement axes;
- S_{max}, computed from the correlation tensor and corresponding to the maximal CHSH value achievable after optimization over measurement settings.

This distinction is physically important because a decrease of S_{fixed} does not necessarily imply degradation of the intrinsic nonlocal properties of the quantum state.

Model and Analytical Predictions

The reference state is the Bell state

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}} \left(|01\rangle + |10\rangle\right).$$

Two different state families are analyzed.

Phase-evolved states.
The first family is generated through fixed local phase evolution:

$$|\psi(\theta)\rangle = \frac{1}{\sqrt{2}} \left(|01\rangle + e^{i\theta}|10\rangle\right), \tag{6.8}$$

obtained by applying the local unitary transformation

$$U_A(\theta) \otimes I_B.$$

Depolarized states.
The second family corresponds to the depolarizing channel:

$$\rho(p) = (1-p)\rho_0 + \frac{p}{4} I_4, \tag{6.9}$$

where

$$\rho_0 = |\Psi^+\rangle\langle\Psi^+|, \qquad p \in [0,1].$$

The fixed CHSH measurement settings are chosen to be optimal for the Bell state at $\theta = 0$, whose correlation tensor is

$$T(0) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

The local analyzer directions for subsystem A are

$$\mathbf{a}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad \mathbf{a}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix},$$

while the analyzer directions for subsystem B are

$$\mathbf{b}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \qquad \mathbf{b}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}.$$

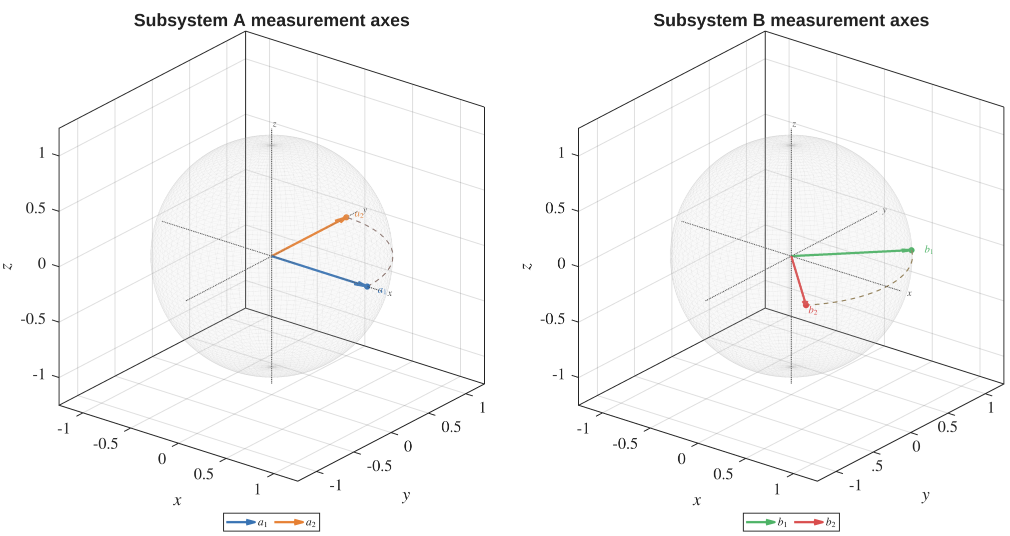


Figure 6.9: Fixed CHSH measurement axes represented on the Bloch spheres of subsystems A and B. The analyzer directions define the fixed laboratory configuration used to evaluate S_{fixed}.

The corresponding Bloch-sphere geometry is shown in Figure 6.9.

With these settings, S_{fixed} represents the CHSH value measured using a fixed analyzer configuration, whereas $S_{\max}$ represents the maximal violation achievable after optimization over local measurement axes.

For the phase-evolved Bell state, the correlation tensor becomes

$$T(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{6.10}$$

Using the fixed analyzer directions introduced above, the corresponding CHSH value is

$$S_{\text{fixed}}(\theta) = 2\sqrt{2}\cos\theta.$$

In the numerical plots, the quantity

$$|S_{\text{fixed}}|$$

is represented, since CHSH violation depends on the condition

$$|S| > 2.$$

The maximal CHSH value is instead independent of the phase parameter:

$$S_{\max} = 2\sqrt{2}.$$

This result reflects the invariance of nonlocality under local unitary transformations.

For the depolarizing channel, the correlation tensor is uniformly contracted:

$$T(p) = (1-p)T(0).$$

Consequently,

$$S_{\text{fixed}}(p) = 2\sqrt{2}(1-p), \qquad S_{\max}(p) = 2\sqrt{2}(1-p).$$

The CHSH violation threshold is determined from

$$S_{\max}(p) > 2,$$

which gives

$$p < p_{\text{CHSH}} = 1 - \frac{1}{\sqrt{2}} \simeq 0.2929.$$

Numerical Results

The numerical results are summarized in Figure 6.10.
In the fixed-phase evolution regime, $|S_{\text{fixed}}|$ oscillates between zero and the Tsirelson bound.
This behavior originates from the relative rotation between the fixed laboratory analyzer axes and the phase-dependent correlation tensor.
At the same time, the maximal CHSH value remains constant:

$$S_{\max} = 2\sqrt{2},$$

demonstrating that the intrinsic nonlocality of the state is fully preserved.

In the depolarizing regime, the numerical and analytical curves overlap within numerical precision.

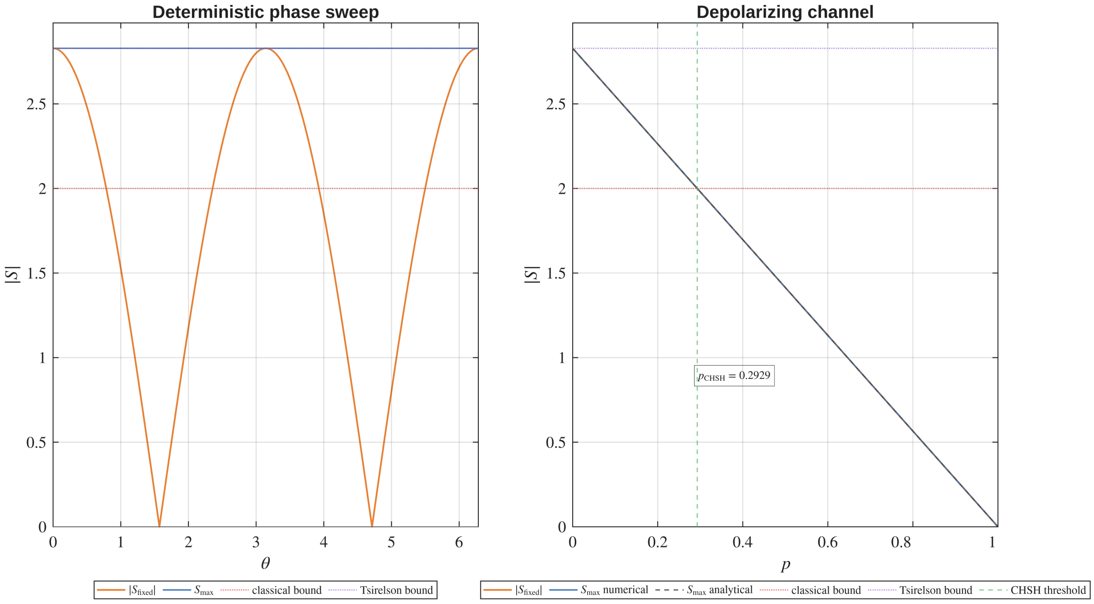


Figure 6.10: CHSH behavior under fixed phase evolution and depolarizing noise. The left panel shows $|S_{\text{fixed}}|$ and S_{max} as functions of the phase parameter θ. The right panel shows the corresponding quantities as functions of the depolarizing parameter p. The red dotted line indicates the classical CHSH bound $|S| = 2$, while the purple dotted line corresponds to the Tsirelson bound $2\sqrt{2}$.

Both $|S_{\text{fixed}}|$ and S_{max} decrease linearly with the depolarization parameter p, crossing the classical CHSH bound at

$$p_{\text{CHSH}} \simeq 0.2929.$$

Unlike the coherent phase case, the reduction of CHSH violation now reflects a genuine loss of nonlocal correlation strength.

Discussion

The two propagation mechanisms considered in this experiment produce qualitatively different effects on CHSH nonlocality.
In the fixed-phase evolution regime, the quantum state remains pure and maximally entangled for all values of the phase parameter θ.
Consequently, the oscillatory behavior of $|S_{\text{fixed}}|$ does not indicate degradation of nonlocality. Instead, it reflects the fact that the fixed laboratory analyzer directions become progressively misaligned with the rotated correlation tensor.
This explains why $|S_{\text{fixed}}|$ may fall below the classical CHSH threshold, or even vanish completely, while the optimized quantity S_{max} remains fixed at the Tsirelson bound.
The apparent loss of CHSH violation is therefore caused entirely by measurement-axis mismatch.

The depolarizing regime exhibits fundamentally different behavior.
The channel reduces the magnitude of the correlation tensor itself, producing a genuine contraction of the available nonlocal correlations.

As a consequence, no optimization of the analyzer directions can recover CHSH violation once the depolarization parameter exceeds the threshold

$$p_{\mathrm{CHSH}} = 1 - \frac{1}{\sqrt{2}}.$$

The experiment therefore establishes an important operational distinction:

- S_{fixed} is sensitive to both measurement misalignment and physical degradation;
- $S_{\max}$ isolates the maximal nonlocality intrinsically present in the quantum state.

Physical Interpretation

This experiment separates coherent fiber-induced transformations from incoherent decoherence mechanisms.
A fixed relative phase acts as a local unitary transformation generated by birefringence-induced phase accumulation between polarization modes.
Such a mechanism rotates the correlation tensor but does not modify its singular values. Consequently, the intrinsic nonlocal resources of the Bell state remain fully preserved.
However, a fixed analyzer configuration may fail to detect the CHSH violation if the laboratory axes are no longer aligned with the rotated tensor.

Depolarization has a fundamentally different physical interpretation.
It models the effective loss of polarization coherence caused by unresolved degrees of freedom, stochastic fluctuations, frequency-dependent propagation effects, or polarization-mode dispersion.
In this case, the correlation tensor is not merely rotated but genuinely contracted, leading to irreversible degradation of the nonlocal correlations.
The comparison between S_{fixed} and $S_{\max}$ therefore provides an operational diagnostic tool:

- if S_{fixed} changes while $S_{\max}$ remains constant, the effect is compatible with coherent unitary evolution;
- if $S_{\max}$ decreases, the quantum state itself has lost nonlocal correlation strength.

For the quantum fiber simulator, this distinction is particularly important.
A fiber-induced phase shift may hide CHSH violation from fixed laboratory measurements without destroying the underlying entanglement resource, whereas depolarizing effects correspond to genuine physical degradation of the entangled photon pair.

6.5 Experiment 5: Composite Fiber Channel

Objective and Physical Motivation

The fifth experiment combines the two main physical mechanisms introduced in the previous experiments: coherent residual phase evolution and local incoherent depolarization.
The objective is to model the optical fiber not only as a fixed-phase shifter, but as an effective composite quantum channel acting independently on the two propagation arms.
The channel is constructed by first applying the residual phase evolution and then applying independent local depolarizing channels on subsystems A and B.
This produces the effective transformation

$$\rho_{AB}^{\text{out}} = (\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}) \left[U_A(\theta) \otimes U_B \, \rho_{AB}^{\text{in}} \, (U_A(\theta) \otimes U_B)^\dagger \right], \tag{6.11}$$

where $U_B = I$ and $U_A(\theta)$ is the reduced phase unitary.
This experiment therefore represents the first complete effective fiber model considered in the simulator, where coherent and incoherent effects are treated within the same density-operator framework.

Model and Analytical Predictions

The input state is the Bell state

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}} \left(|01\rangle + |10\rangle \right).$$

After the local phase evolution, the state becomes

$$|\psi(\theta)\rangle = \frac{1}{\sqrt{2}} \left(|01\rangle + e^{i\theta} |10\rangle \right).$$

The two-arm depolarizing channel is then applied with independent depolarization parameters p_A and p_B.
For two-body Pauli correlations, local depolarization produces the contraction factor

$$\eta = (1 - p_A)(1 - p_B). \tag{6.12}$$

Therefore, in the pure single-realization branch, the diagonal correlations are

$$c_{xx} = \eta \cos\theta, \qquad c_{yy} = \eta \cos\theta, \qquad c_{zz} = -\eta.$$

The maximal CHSH parameter is predicted from the singular values of the correlation tensor. Since local depolarization contracts the correlation tensor by η, the maximal CHSH value is reduced accordingly:

$$S_{\max} = 2\sqrt{2}\,\eta.$$

For the statistical ensemble extension, phase realizations θ_k are sampled and averaged after the composite channel is applied to each realization:

$$\rho_{AB}^{\mathrm{ens}} = \frac{1}{N}\sum_{k=1}^{N} (\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}) \left[U_A(\theta_k) \otimes U_B \, \rho_{AB}^{\mathrm{in}} \, (U_A(\theta_k) \otimes U_B)^\dagger \right]. \tag{6.13}$$

Introducing again the ensemble coherence parameter

$$\mu = \langle e^{i\theta} \rangle,$$

the ensemble-averaged transverse correlations become

$$c_{xx} = \eta \,\mathrm{Re}(\mu), \qquad c_{yy} = \eta \,\mathrm{Re}(\mu), \qquad c_{zz} = -\eta.$$

Thus, the composite model separates two distinct degradation mechanisms:

- μ describes coherence loss due to unresolved phase fluctuations;
- η describes correlation contraction due to independent local depolarization.

Numerical Results

The experiment is divided into two complementary parts.
First, the pure-branch composite channel is evaluated over a fixed sweep of θ. This isolates the effect of local depolarization on a single phase-evolved realization.

Figure 6.11 shows the pure-channel summary.

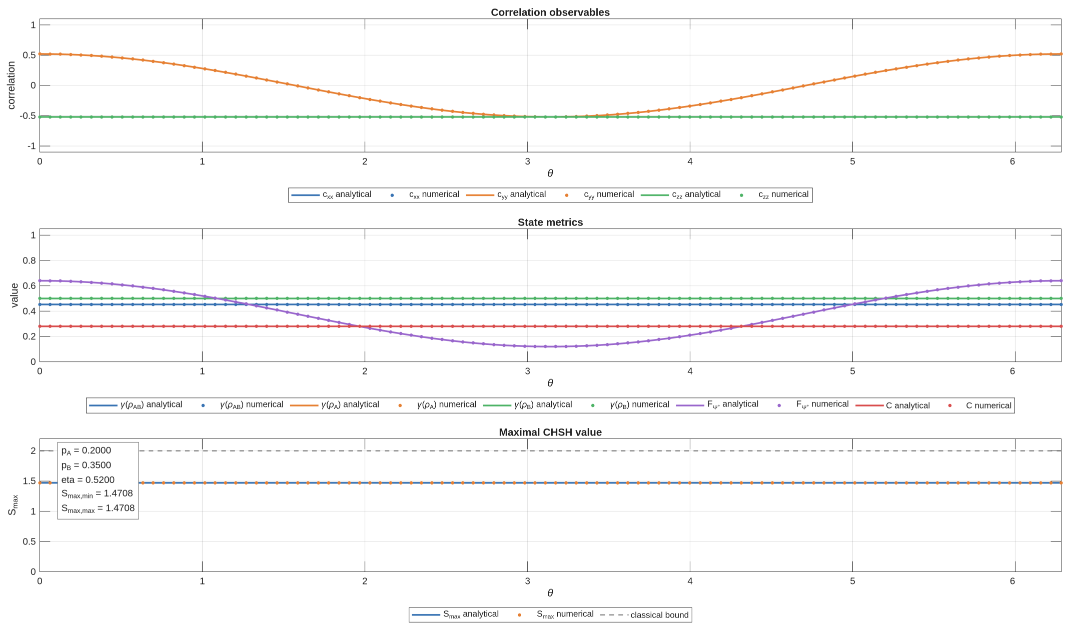


Figure 6.11: Pure composite fiber channel: local depolarization contracts the phase-dependent correlations and lowers the maximal CHSH value, while the analytical and numerical curves remain indistinguishable at the plotted scale.

The transverse correlations c_{xx} and c_{yy} retain their sinusoidal dependence on θ, but their amplitude is reduced by the contraction factor η. The longitudinal correlation c_{zz} remains independent of θ, but is also contracted by the same factor.

The second part introduces statistical phase ensembles. Three representative distributions are considered:

- constant phase distribution;
- uniform phase distribution;
- Gaussian phase distribution.

Figure 6.12 shows the constant-phase ensemble.

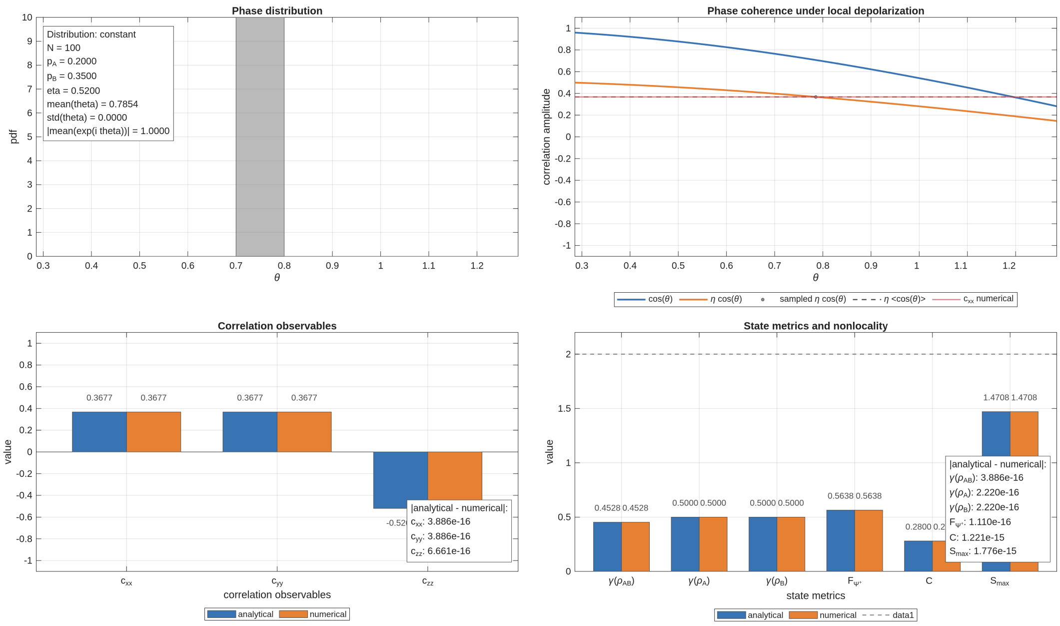


Figure 6.12: Composite channel with constant phase distribution: the ensemble reduces to a single phase realization, while local depolarization reduces purity, fidelity, concurrence, and nonlocality.

In this regime, the stochastic part of the model is inactive, and the observed degradation is entirely due to local depolarization.

Figure 6.13 shows the uniform phase distribution.
The transverse coherence is suppressed by the phase average, while the depolarizing channel reduces the remaining correlation structure.

Figure 6.14 shows the Gaussian phase distribution.
This case represents an intermediate propagation regime, where neither phase coherence nor entanglement is completely suppressed.

The corresponding correlation tensor snapshots are shown in Figure 6.15.
The tensor snapshots confirm that the composite model reproduces the expected analytical structure across all tested phase distributions.

Discussion

The numerical results show that phase averaging and local depolarization affect the state in distinct but complementary ways.
The phase distribution controls the coherence parameter μ. When the distribution is narrow or concentrated around a single phase value, coherence is preserved. When the distribution becomes broad, transverse coherence is suppressed.

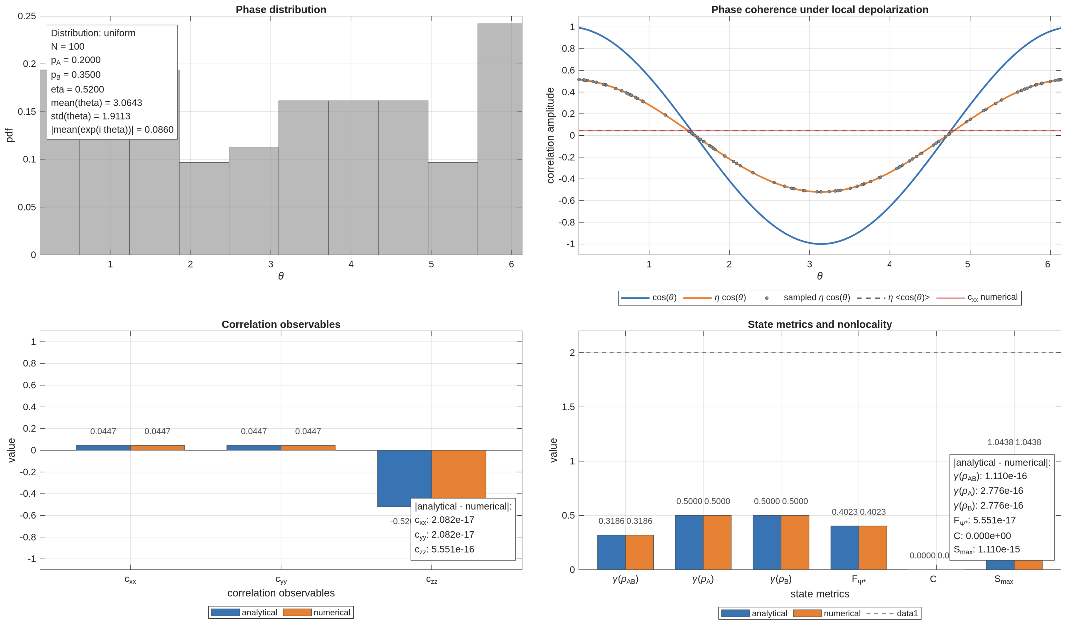


Figure 6.13: Composite channel with uniform phase distribution: phase averaging suppresses transverse coherence, while local depolarization further contracts the remaining longitudinal correlation.

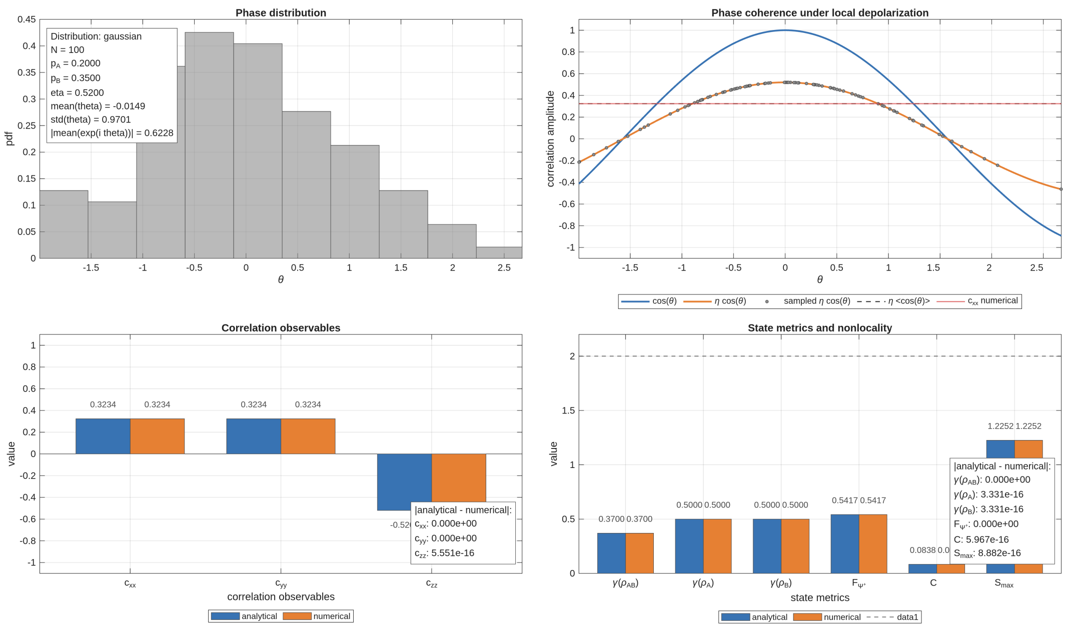
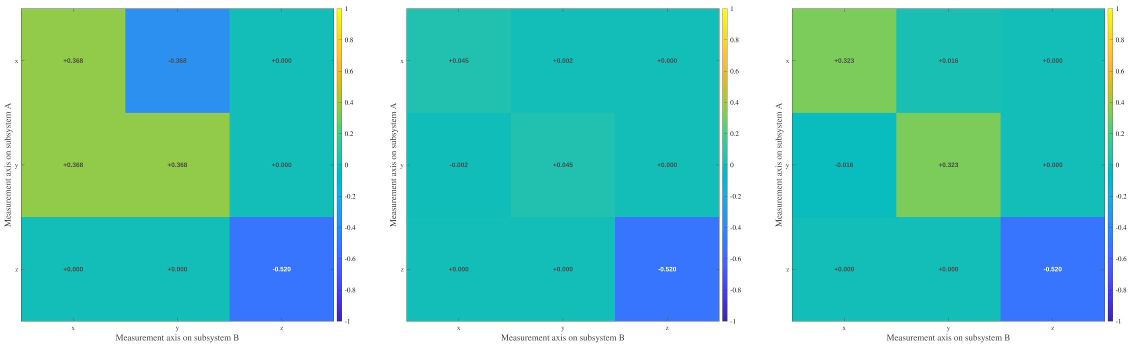


Figure 6.14: Composite channel with Gaussian phase distribution: partial phase coherence survives ensemble averaging, producing intermediate values of purity, fidelity, concurrence, and maximal CHSH parameter.



(a) Constant phase (b) Uniform distribution (c) Gaussian distribution

Figure 6.15: Correlation tensor snapshots for the composite fiber channel. The tensor structure reflects the combined action of statistical phase averaging and local two-arm depolarization: phase fluctuations suppress transverse coherence, while local depolarization contracts all two-body correlations by the factor η.

Local depolarization instead acts through the contraction factor η, reducing the magnitude of two-body correlations independently of the particular phase realization.
The composite model therefore produces the effective structure

$$\text{coherent phase behavior} \quad \longrightarrow \quad \mu \quad \longrightarrow \quad \eta\mu,$$

for the transverse correlation terms.
This separation is physically useful because it distinguishes two mechanisms that may otherwise appear similar at the level of measured correlations:

- loss of phase coherence due to unresolved stochastic phase evolution;
- local polarization degradation due to depolarizing effects in each fiber arm.

In the pure-branch case, the phase remains fixed, and the state degradation is entirely due to local depolarization.
In the ensemble case, both mechanisms are present. The resulting decrease of purity, fidelity, concurrence, and maximal CHSH value reflects the combined effect of coherent uncertainty and incoherent local noise.

Physical Interpretation

The composite channel provides a more realistic effective description of fiber propagation than the previous isolated models.
The residual phase θ describes a coherent birefringent effect that rotates the correlation tensor in the transverse plane. If this phase is fixed, the evolution remains coherent and can in principle be compensated.
The depolarizing parameters p_A and p_B, instead, describe incoherent degradation in the two independent fiber arms. Their effect cannot be represented as a unitary rotation, because they reduce the magnitude of the polarization correlations.
The contraction factor η encodes the fact that two-body correlations survive only to the extent that polarization information is preserved in both arms.

When phase fluctuations are also present, the ensemble coherence parameter μ determines how much of the coherent transverse structure remains after averaging. The composite fiber model therefore captures the simultaneous presence of:

- fixed or stochastic coherent phase evolution;
- independent local depolarization in the two arms;
- reduction of Bell-state fidelity;
- reduction of concurrence;
- degradation of CHSH nonlocality.

This experiment represents the final step of the present numerical validation pipeline: it connects the reduced phase model, random phase ensembles, depolarizing channels, and CHSH diagnostics into a single effective fiber-channel description.

Appendix A

Foundations of Bipartite Quantum States

A.1 Explicit Derivation of Reduced Density Operators and Comparison of Global States

This appendix provides an explicit derivation of the reduced density operators associated with the phase-evolved Bell state introduced in Chapter 3. It also illustrates an important conceptual point: identical local reduced states may correspond to physically different global states.

The discussion clarifies why local measurements alone are insufficient to characterize bipartite quantum systems and motivates the introduction of global quantities such as purity, correlation functions, and entanglement measures.

Phase-Evolved Bell State

We consider the two-qubit state

$$|\psi(\theta)\rangle = \frac{|01\rangle + e^{i\theta}|10\rangle}{\sqrt{2}}, \tag{A.1}$$

written in the computational basis

$$\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}.$$

Its vector representation is

$$|\psi(\theta)\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ e^{i\theta} \\ 0 \end{pmatrix}. \tag{A.2}$$

The corresponding density operator is

$$\rho_{AB}(\theta) = |\psi(\theta)\rangle\langle\psi(\theta)|. \tag{A.3}$$

Expanding explicitly gives

$$\rho_{AB}(\theta) = \frac{1}{2}\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & e^{-i\theta} & 0 \\ 0 & e^{i\theta} & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{A.4}$$

Explicit Evaluation of the Partial Trace

The reduced density operator of subsystem A is obtained through the partial trace over subsystem B:

$$\rho_A(\theta) = \mathrm{Tr}_B\left(\rho_{AB}(\theta)\right). \tag{A.5}$$

Using the computational basis of subsystem B,

$$\{|0\rangle, |1\rangle\},$$

the partial trace can be written explicitly as

$$\rho_A(\theta) = \sum_{b=0}^{1} \left(I_A \otimes \langle b|\right) \rho_{AB}(\theta) \left(I_A \otimes |b\rangle\right). \tag{A.6}$$

We now evaluate the individual contributions.

The diagonal terms give

$$\begin{aligned} \mathrm{Tr}_B\left(|01\rangle\langle 01|\right) &= |0\rangle\langle 0|, \\ \mathrm{Tr}_B\left(|10\rangle\langle 10|\right) &= |1\rangle\langle 1|. \end{aligned} \tag{A.7}$$

The off-diagonal terms vanish:

$$\begin{aligned} \mathrm{Tr}_B\left(|01\rangle\langle 10|\right) &= 0, \\ \mathrm{Tr}_B\left(|10\rangle\langle 01|\right) &= 0, \end{aligned} \tag{A.8}$$

due to the orthogonality relation

$$\langle 0|1\rangle = 0.$$

Summing all contributions yields

$$\rho_A(\theta) = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{I}{2}. \tag{A.9}$$

By symmetry,

$$\rho_B(\theta) = \frac{I}{2}. \tag{A.10}$$

Thus, although the global state depends explicitly on the phase θ, the reduced states do not. The phase appears only in the off-diagonal coherence terms of the global density operator, which are eliminated by the partial trace.

In particular, for $\theta = 0$,

$$|\Psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}, \tag{A.11}$$

one obtains the same reduced states,

$$\rho_A = \rho_B = \frac{I}{2}. \tag{A.12}$$

Local Equivalence and Global Distinction

The previous result illustrates an important conceptual property of bipartite quantum systems.

Consider the classical mixture

$$\rho^{(\text{class})} = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{I}{2}. \tag{A.13}$$

At the local level,

$$\rho_A^{(\text{ent})} = \rho^{(\text{class})}, \tag{A.14}$$

and therefore, for any observable O_A,

$$\mathrm{Tr}\left(\rho^{(\text{class})} O_A\right) = \mathrm{Tr}\left(\rho_A^{(\text{ent})} O_A\right). \tag{A.15}$$

Hence, local measurements cannot distinguish classical mixtures from reductions of entangled states.

The distinction emerges only at the global level.
Consider the separable mixed state

$$\rho_{AB}^{(\text{sep})} = \frac{1}{2}|01\rangle\langle 01| + \frac{1}{2}|10\rangle\langle 10|. \tag{A.16}$$

Its reduced states are identical to those of the entangled Bell state:

$$\begin{aligned} \mathrm{Tr}_B\left(\rho_{AB}^{(\text{sep})}\right) &= \frac{I}{2}, \\ \mathrm{Tr}_A\left(\rho_{AB}^{(\text{sep})}\right) &= \frac{I}{2}. \end{aligned} \tag{A.17}$$

However, the separable state lacks the coherence terms present in the entangled density operator.

This difference becomes visible through global quantities.
For the entangled state,

$$\gamma\left(\rho_{AB}(\theta)\right) = 1, \tag{A.18}$$

whereas for the separable mixture,

$$\gamma\left(\rho_{AB}^{(\text{sep})}\right) = \frac{1}{2}. \tag{A.19}$$

A second distinction appears in bipartite correlations.

For the entangled phase-evolved state,

$$\begin{aligned}\langle \sigma_x \otimes \sigma_x \rangle &= \cos\theta, \\ \langle \sigma_y \otimes \sigma_y \rangle &= \cos\theta,\end{aligned} \tag{A.20}$$

whereas for the separable mixture,

$$\begin{aligned}\langle \sigma_x \otimes \sigma_x \rangle &= 0, \\ \langle \sigma_y \otimes \sigma_y \rangle &= 0.\end{aligned} \tag{A.21}$$

Thus, states with identical reduced density operators may differ fundamentally at the global level.
Reduced states encode all locally accessible information, but do not capture global coherence or bipartite entanglement.

A.2 Derivation of Fidelity for a Pure Reference State

This appendix derives the simplified expression for quantum fidelity when one of the two states is pure. The result is used throughout Chapter 4 in the evaluation of fidelity with respect to Bell states.

Beyond its algebraic simplification, this derivation also clarifies the operational interpretation of fidelity as a projection probability onto a target quantum state.

General Definition of Fidelity

Let ρ and σ be density operators acting on a finite-dimensional Hilbert space.
The general definition of quantum fidelity is

$$F(\rho, \sigma) = \left(\mathrm{Tr} \sqrt{\sqrt{\rho}\, \sigma \sqrt{\rho}} \right)^2 . \tag{A.22}$$

In the present derivation, we consider the special case in which the reference state is pure:

$$\sigma = |\psi\rangle\langle\psi|, \tag{A.23}$$

with

$$\langle\psi|\psi\rangle = 1. \tag{A.24}$$

By construction, the operator σ is:

- Hermitian,
$$\sigma^\dagger = \sigma,$$
- positive semidefinite,
$$\langle\chi|\sigma|\chi\rangle \geq 0, \qquad \forall\, |\chi\rangle,$$
- normalized,
$$\mathrm{Tr}(\sigma) = 1.$$

Moreover, σ is a rank-one projector onto the one-dimensional subspace spanned by $|\psi\rangle$.

Reduction for a Pure Reference State

Substituting Eq. (A.23) into Eq. (A.22) gives

$$\sqrt{\rho}\,\sigma\,\sqrt{\rho} = \sqrt{\rho}\,|\psi\rangle\langle\psi|\,\sqrt{\rho}. \tag{A.25}$$

Define the vector

$$|\phi\rangle = \sqrt{\rho}\,|\psi\rangle. \tag{A.26}$$

Then Eq. (A.25) becomes

$$\sqrt{\rho}\,\sigma\,\sqrt{\rho} = |\phi\rangle\langle\phi|. \tag{A.27}$$

The operator $|\phi\rangle\langle\phi|$ is again rank one and positive semidefinite.

Spectral Structure of the Rank-One Operator

The only nonzero eigenvalue of the operator in Eq. (A.27) is

$$\lambda = \langle\phi|\phi\rangle. \tag{A.28}$$

Introducing the normalized vector

$$|\hat{\phi}\rangle = \frac{|\phi\rangle}{\|\phi\|}, \tag{A.29}$$

the spectral decomposition becomes

$$|\phi\rangle\langle\phi| = \lambda\,|\hat{\phi}\rangle\langle\hat{\phi}|. \tag{A.30}$$

The square root of the operator is therefore

$$\sqrt{|\phi\rangle\langle\phi|} = \sqrt{\lambda}\,|\hat{\phi}\rangle\langle\hat{\phi}|. \tag{A.31}$$

Taking the trace gives

$$\mathrm{Tr}\,\sqrt{|\phi\rangle\langle\phi|} = \sqrt{\lambda}. \tag{A.32}$$

Substituting into Eq. (A.22), one obtains

$$F(\rho,\sigma) = \lambda = \langle\phi|\phi\rangle. \tag{A.33}$$

Using Eq. (A.26),

$$\lambda = \langle\psi|\rho|\psi\rangle. \tag{A.34}$$

Final Expression and Interpretation

Combining the previous results, the fidelity simplifies to

$$F(\rho, |\psi\rangle\langle\psi|) = \langle\psi|\rho|\psi\rangle. \tag{A.35}$$

This is the expression used throughout Chapter 4 for fidelity with respect to Bell states.

The result admits a direct operational interpretation. Since

$$|\psi\rangle\langle\psi|$$

is the projector onto the state $|\psi\rangle$, the fidelity corresponds to the probability of obtaining the outcome $|\psi\rangle$ in a projective measurement performed on the state ρ.

Fidelity therefore quantifies the overlap between the physical state produced by the system and the chosen target reference state.

A.3 Derivations for Concurrence and Reduced Density Operators

This appendix derives the relations connecting concurrence, reduced density operators, and subsystem purity for pure bipartite two-qubit states.

The derivation clarifies how bipartite entanglement is encoded in the mixedness of the reduced subsystems, providing the theoretical foundation for the relations used throughout Chapter 4.

Coefficient-Matrix Representation

Consider a normalized two-qubit pure state written in the computational basis as

$$|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle. \tag{A.36}$$

Introduce the coefficient matrix

$$A_\psi = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \tag{A.37}$$

so that the state can be expressed compactly as

$$|\psi\rangle = \sum_{i,j=0}^{1} (A_\psi)_{ij}\, |i\rangle \otimes |j\rangle. \tag{A.38}$$

This matrix representation provides a convenient bridge between the bipartite wavefunction and the reduced density operators.

Reduced Density Operators

The global density operator is

$$\rho_{AB} = |\psi\rangle\langle\psi|. \tag{A.39}$$

Expanding Eq. (A.38) yields

$$\rho_{AB} = \sum_{i,j,k,l} (A_\psi)_{ij}(A_\psi)^*_{kl}\, |i\rangle\langle k| \otimes |j\rangle\langle l|. \tag{A.40}$$

The reduced density operator on subsystem A is defined by the partial trace

$$\rho_A = \mathrm{Tr}_B(\rho_{AB}). \tag{A.41}$$

Applying the partial trace to Eq. (A.40) gives

$$\rho_A = \sum_{i,j,k,l} (A_\psi)_{ij}(A_\psi)^*_{kl} \left(\sum_v \langle v|j\rangle\langle l|v\rangle \right) |i\rangle\langle k|. \tag{A.42}$$

Using the orthogonality relation

$$\sum_v \langle v|j\rangle\langle l|v\rangle = \delta_{jl}, \tag{A.43}$$

one obtains

$$\rho_A = \sum_{i,k,j} (A_\psi)_{ij}(A_\psi)^*_{kj}\, |i\rangle\langle k|. \tag{A.44}$$

Therefore, in matrix form,

$$\rho_A = A_\psi A_\psi^\dagger. \tag{A.45}$$

Similarly,

$$\rho_B = A_\psi^\dagger A_\psi. \tag{A.46}$$

Pure-State Concurrence

For pure two-qubit states, concurrence is defined as

$$C(|\psi\rangle) = |\langle\psi|(\sigma_y \otimes \sigma_y)|\psi^*\rangle|\,. \tag{A.47}$$

Using the computational-basis representation,

$$|\psi\rangle = \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}, \qquad |\psi^*\rangle = \begin{pmatrix} a^* \\ b^* \\ c^* \\ d^* \end{pmatrix}, \tag{A.48}$$

one finds

$$(\sigma_y \otimes \sigma_y)|\psi^*\rangle = \begin{pmatrix} -d^* \\ c^* \\ b^* \\ -a^* \end{pmatrix}. \tag{A.49}$$

Taking the inner product yields

$$\langle\psi|(\sigma_y \otimes \sigma_y)|\psi^*\rangle = 2(ad - bc)^*. \tag{A.50}$$

Therefore,

$$C(|\psi\rangle) = 2|ad - bc|. \tag{A.51}$$

Using

$$\det(A_\psi) = ad - bc, \tag{A.52}$$

the concurrence becomes

$$C(|\psi\rangle) = 2|\det(A_\psi)|. \tag{A.53}$$

Relation with Reduced Density Operators

Using Eq. (A.45),

$$\rho_A = A_\psi A_\psi^\dagger, \tag{A.54}$$

one obtains

$$\det(\rho_A) = \det(A_\psi)\det(A_\psi^\dagger) = |\det(A_\psi)|^2. \tag{A.55}$$

Substituting into Eq. (A.53) gives

$$C(|\psi\rangle) = 2\sqrt{\det(\rho_A)}. \tag{A.56}$$

This reproduces the relation introduced in Chapter 4 [see Eq. (4.18)].

An identical relation holds for subsystem B:

$$C(|\psi\rangle) = 2\sqrt{\det(\rho_B)}. \tag{A.57}$$

These expressions show that bipartite entanglement is directly encoded in the reduced density operators.

Relation with Purity

For a single-qubit density operator ρ, one has

$$\mathrm{Tr}(\rho) = 1, \tag{A.58}$$

and purity

$$\gamma(\rho) = \mathrm{Tr}(\rho^2). \tag{A.59}$$

For 2×2 density operators, the determinant satisfies

$$\det(\rho) = \frac{1 - \gamma(\rho)}{2}. \tag{A.60}$$

Substituting Eq. (A.60) into Eq. (4.18) yields

$$C(|\psi\rangle) = \sqrt{2\,(1 - \gamma(\rho_A))}. \tag{A.61}$$

Therefore, for pure bipartite states, the mixedness of the reduced subsystem provides a direct quantitative measure of entanglement.

Separable states produce pure reduced density operators and therefore zero concurrence, whereas maximally entangled states produce maximally mixed reduced states and maximal concurrence.

Appendix B

Basis Representation and Invariance Properties

B.1 Concurrence: Basis Dependence, Representation Dependence, and Invariance

This appendix clarifies the distinction between representation-dependent quantities and physically invariant quantities in quantum mechanics, with particular emphasis on the concurrence used throughout the present work.

A potential conceptual ambiguity arises from the standard definition of concurrence for mixed two-qubit states, since the construction explicitly involves the operator σ_y and complex conjugation in the computational basis. At first sight, this may suggest that concurrence depends on a preferred polarization axis or on a specific basis representation.
The purpose of this appendix is to show that this interpretation is incorrect, and that concurrence constitutes an intrinsic, basis-independent measure of bipartite entanglement.

Representation Dependence and Basis Transformations

In quantum mechanics, vectors and operators are represented through coordinates and matrices relative to a chosen basis.
For a single qubit, a generic pure state may be written as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle.$$

The coefficients α and β depend on the chosen basis representation.
For example, switching from the computational basis

$$\{|0\rangle, |1\rangle\}$$

to the rotated basis

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}},$$

changes the numerical coordinates of the state vector, while leaving the physical quantum state unchanged.

Similarly, a density operator transforms under a basis change according to

$$\rho \to \rho' = U \rho U^\dagger, \tag{B.1}$$

where U is a unitary change-of-basis operator.
The individual matrix entries of ρ therefore depend on the chosen representation.
However, physically meaningful invariant quantities such as

$$\mathrm{Tr}(\rho), \qquad \mathrm{Tr}(\rho^2), \qquad \det(\rho), \qquad \mathrm{spec}(\rho)$$

remain unchanged under Eq. (B.1).

This distinction is fundamental:

- matrix elements and coordinate representations are generally basis dependent,
- intrinsic physical properties are basis independent.

Direction Dependence and Physical Observables

A separate concept is the dependence of observables on physical measurement directions.
Expectation values such as

$$\langle \sigma_x \rangle, \qquad \langle \sigma_y \rangle, \qquad \langle \sigma_z \rangle$$

correspond to measurements performed along different axes of the Bloch sphere.
These quantities are physically direction dependent, since changing the measurement axis changes the measured observable.

For instance,

$$\langle \sigma_z \rangle = \mathrm{Tr}(\rho \sigma_z)$$

measures polarization along the computational Z-axis, while

$$\langle \sigma_x \rangle = \mathrm{Tr}(\rho \sigma_x)$$

measures polarization along the transverse X-axis.

Direction dependence therefore refers to the physical choice of measurement axis, rather than to the mathematical choice of representation basis.
These two notions are related but conceptually distinct.

Representation Dependence of Pauli Matrices

The Pauli operators themselves admit different matrix representations depending on the chosen basis.
In the computational basis,

$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}. \tag{B.2}$$

Under a basis transformation U, the matrix representation changes according to

$$\sigma_y \to \sigma'_y = U\sigma_y U^\dagger. \tag{B.3}$$

Consequently, the matrix entries appearing in Eq. (B.2) are representation dependent.

Importantly, this does not imply that the physical observable itself changes. Rather, only its coordinate representation changes.
This situation is directly analogous to ordinary vector coordinates in Euclidean geometry.

Concurrence and the Spin-Flip Construction

For mixed two-qubit states, concurrence is commonly expressed through the spin-flipped density operator

$$\tilde{\rho} = (\sigma_y \otimes \sigma_y)\rho^*(\sigma_y \otimes \sigma_y), \tag{B.4}$$

where ρ^* denotes complex conjugation in the computational basis.
The concurrence is then defined as

$$C(\rho) = \max\left(0, \lambda_1 - \lambda_2 - \lambda_3 - \lambda_4\right), \tag{B.5}$$

where λ_i are the ordered square roots of the eigenvalues of

$$\rho\tilde{\rho}.$$

At first sight, Eq. (B.4) may appear to privilege the Y-direction of the Bloch sphere because of the explicit presence of σ_y.
However, in this context, σ_y is not used as a measurement operator associated with the physical Y-axis. Instead, it appears because it realizes the invariant antisymmetric structure of two-dimensional spinor spaces.

Antisymmetric Structure of Two-Qubit States

Consider a generic pure two-qubit state

$$|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle. \tag{B.6}$$

The coefficients may be organized into the matrix

$$\Psi = \begin{pmatrix} a & b \\ c & d \end{pmatrix}. \tag{B.7}$$

For pure states, concurrence reduces to

$$C(|\psi\rangle) = 2|ad - bc|. \tag{B.8}$$

The quantity

$$ad - bc$$

is precisely the determinant of the coefficient matrix Ψ:

$$\det(\Psi) = ad - bc. \tag{B.9}$$

This determinant originates from the antisymmetric bilinear form

$$\varepsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \tag{B.10}$$

which satisfies

$$\sigma_y = -i\varepsilon. \tag{B.11}$$

Thus, the appearance of σ_y in the concurrence construction is not associated with a preferred physical polarization axis, but with the antisymmetric invariant structure of two-level quantum systems.

Basis Independence of Concurrence

An entanglement measure must remain invariant under local unitary transformations, since local polarization rotations cannot create or destroy entanglement.
For any local unitary evolution,

$$\rho \to (U_A \otimes U_B)\rho(U_A^\dagger \otimes U_B^\dagger), \tag{B.12}$$

the concurrence satisfies

$$C\left[(U_A \otimes U_B)\rho(U_A^\dagger \otimes U_B^\dagger)\right] = C(\rho). \tag{B.13}$$

This invariance is essential for the physical interpretation of concurrence as an intrinsic measure of bipartite entanglement.

Consequently:

- the matrix representation of the spin-flipped operator depends on the chosen basis,
- the intermediate construction in Eq. (B.4) is representation dependent,
- the final concurrence $C(\rho)$ is basis independent.

Manifestly Basis-Independent Formulations

For pure states, concurrence admits formulations that make the invariance more transparent.
Using the reduced density operator

$$\rho_A = \mathrm{Tr_B}\left(|\psi\rangle\langle\psi|\right),$$

one obtains

$$C(|\psi\rangle) = 2\sqrt{\det(\rho_A)}. \tag{B.14}$$

Equivalently,

$$C(|\psi\rangle) = \sqrt{2\,(1 - \mathrm{Tr}(\rho_A^2))}. \tag{B.15}$$

These expressions involve only invariant quantities such as determinants and traces, and therefore do not rely explicitly on any preferred basis representation.

More fundamentally, concurrence for mixed states is defined through the convex-roof construction

$$C(\rho) = \min_{\{p_k, |\psi_k\rangle\}} \sum_k p_k C(|\psi_k\rangle), \tag{B.16}$$

where

$$\rho = \sum_k p_k |\psi_k\rangle\langle\psi_k|. \tag{B.17}$$

This formulation is intrinsically basis independent.
The spin-flip formalism of Eq. (B.4) constitutes a special analytical construction that allows the explicit evaluation of the convex-roof optimization for two-qubit systems.

Physical Interpretation in Fiber-Based Quantum Channels

The distinction between representation-dependent quantities and invariant quantities plays a central role in the modeling strategy developed throughout the present work.

Local unitary transformations induced by fiber propagation,

$$U_A \otimes U_B,$$

may alter:

- local Bloch coordinates,
- Stokes parameters,
- tensor components,
- fixed-basis Bell-state fidelities,
- correlation functions evaluated along specific axes.

However, these transformations do not modify the intrinsic entanglement content of the bipartite state.
Consequently, the concurrence remains unchanged under coherent local polarization evolution.

By contrast, non-unitary effects such as ensemble averaging, decoherence, or polarization-mode dispersion generate effective mixed states and therefore reduce concurrence.
This distinction establishes the conceptual transition

$$\text{coherent local evolution} \rightarrow \text{effective decoherence} \rightarrow \text{entanglement degradation},$$

which constitutes one of the central physical mechanisms investigated in the simulator developed in this work. “‘

Appendix C

Quantum Channels and Effective Fiber Models

C.1 Global Depolarizing Channel Analytics

This appendix derives the analytical expressions used to validate the depolarizing-channel experiment in Section ??.
The reference state is the Bell state

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle + |10\rangle\right),$$

with density operator

$$\rho_0 = |\Psi^+\rangle\langle\Psi^+|.$$

The depolarized family is defined as

$$\rho(p) = (1-p)\rho_0 + \frac{p}{4} I_4, \qquad p \in [0,1]. \tag{C.1}$$

Density matrix representation

In the computational basis

$$\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\},$$

the Bell-state density operator is

$$\rho_0 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Substituting this expression into Eq. (C.1) gives

$$\rho(p) = \begin{pmatrix} \frac{p}{4} & 0 & 0 & 0 \\ 0 & \frac{1}{2}-\frac{p}{4} & \frac{1-p}{2} & 0 \\ 0 & \frac{1-p}{2} & \frac{1}{2}-\frac{p}{4} & 0 \\ 0 & 0 & 0 & \frac{p}{4} \end{pmatrix}.$$

Global purity

The global purity is defined as

$$\gamma_{AB}(p) = \mathrm{Tr}\left(\rho(p)^2\right).$$

Since ρ_0 is a rank-one projector and $I_4/4$ is proportional to the identity, the Bell-state direction remains an eigenvector of $\rho(p)$, with eigenvalue

$$\lambda_1 = 1 - \frac{3p}{4}.$$

The three orthogonal Bell-basis directions have eigenvalues

$$\lambda_2 = \lambda_3 = \lambda_4 = \frac{p}{4}.$$

Therefore,

$$\gamma_{AB}(p) = \left(1 - \frac{3p}{4}\right)^2 + 3\left(\frac{p}{4}\right)^2.$$

After simplification,

$$\gamma_{AB}(p) = 1 - \frac{3}{2}p + \frac{3}{4}p^2. \tag{C.2}$$

Reduced density operators

For the Bell state,

$$\mathrm{Tr}_B(\rho_0) = \mathrm{Tr}_A(\rho_0) = \frac{I_2}{2}.$$

Moreover,

$$\mathrm{Tr}_B\left(\frac{I_4}{4}\right) = \mathrm{Tr}_A\left(\frac{I_4}{4}\right) = \frac{I_2}{2}.$$

By linearity of the partial trace,

$$\rho_A(p) = \mathrm{Tr}_B(\rho(p)) = (1-p)\frac{I_2}{2} + p\frac{I_2}{2} = \frac{I_2}{2}.$$

Similarly,

$$\rho_B(p) = \frac{I_2}{2}.$$

The reduced purities are therefore

$$\gamma_A(p) = \gamma_B(p) = \mathrm{Tr}\left[\left(\frac{I_2}{2}\right)^2\right] = \frac{1}{2}. \tag{C.3}$$

Fidelity with respect to $|\Psi^+\rangle$

Since the reference state is pure, the fidelity is

$$F_{\Psi^+}(p) = \langle \Psi^+|\rho(p)|\Psi^+\rangle.$$

Substituting Eq. (C.1),

$$F_{\Psi^+}(p) = (1-p)\langle \Psi^+|\rho_0|\Psi^+\rangle + \frac{p}{4}\langle \Psi^+|I_4|\Psi^+\rangle.$$

Since

$$\langle \Psi^+|\rho_0|\Psi^+\rangle = 1, \qquad \langle \Psi^+|I_4|\Psi^+\rangle = 1,$$

one obtains

$$F_{\Psi^+}(p) = 1 - \frac{3}{4}p. \tag{C.4}$$

Concurrence

The density operator $\rho(p)$ is Bell-diagonal. For a Bell-diagonal two-qubit state, the concurrence is

$$C(p) = \max\left(0, 2\lambda_{\max} - 1\right),$$

where $\lambda_{\max}$ is the largest Bell-basis eigenvalue.
In the present case,

$$\lambda_{\max} = 1 - \frac{3p}{4}.$$

Therefore,

$$C(p) = \max\left(0, 1 - \frac{3p}{2}\right). \tag{C.5}$$

The concurrence vanishes when

$$1 - \frac{3p}{2} = 0,$$

which gives the separability threshold

$$p = \frac{2}{3}. \tag{C.6}$$

Correlation tensor

The Pauli correlations are defined as

$$c_{ij}(p) = \mathrm{Tr}\left[\rho(p)\left(\sigma_i \otimes \sigma_j\right)\right], \qquad i, j \in \{x, y, z\}.$$

Using linearity,

$$c_{ij}(p) = (1-p)\,\mathrm{Tr}\left[\rho_0\left(\sigma_i \otimes \sigma_j\right)\right] + \frac{p}{4}\,\mathrm{Tr}\left[I_4\left(\sigma_i \otimes \sigma_j\right)\right].$$

Since every Pauli product $\sigma_i \otimes \sigma_j$ with $i, j \in \{x, y, z\}$ is traceless,

$$\mathrm{Tr}\left[I_4 \left(\sigma_i \otimes \sigma_j\right)\right] = 0.$$

Thus,

$$c_{ij}(p) = (1-p)c_{ij}(0).$$

Equivalently, the full correlation tensor is uniformly contracted:

$$T(p) = (1-p)T_{\Psi^+}. \tag{C.7}$$

For the reference Bell state $|\Psi^+\rangle$,

$$T_{\Psi^+} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

Therefore,

$$T(p) = \begin{pmatrix} 1-p & 0 & 0 \\ 0 & 1-p & 0 \\ 0 & 0 & -(1-p) \end{pmatrix}. \tag{C.8}$$

The diagonal correlations are

$$c_{xx}(p) = 1-p, \qquad c_{yy}(p) = 1-p, \qquad c_{zz}(p) = -(1-p). \tag{C.9}$$

Summary

Collecting the analytical expressions:

$$\boxed{\begin{aligned} \gamma_{AB}(p) &= 1 - \frac{3}{2}p + \frac{3}{4}p^2, \\ \gamma_A(p) = \gamma_B(p) &= \frac{1}{2}, \\ F_{\Psi^+}(p) &= 1 - \frac{3}{4}p, \\ C(p) &= \max\left(0, 1 - \frac{3}{2}p\right), \\ c_{xx}(p) &= 1-p, \qquad c_{yy}(p) = 1-p, \qquad c_{zz}(p) = -(1-p). \end{aligned}}$$

These expressions provide the analytical reference used for the numerical validation of the depolarizing-channel experiment.

C.2 Commutation of Local Phase Evolution and Isotropic Local Depolarization

This appendix establishes explicitly the commutation relation underlying the composite phase–depolarization fiber model introduced in Section 5.4.

In particular, it is shown that the reduced phase-evolution model considered in this work commutes with isotropic local depolarizing channels acting independently on the two subsystems. This property justifies the equivalence between applying local depolarization before or after coherent phase evolution within the isotropic approximation adopted in the present work.

Effective Channel Structure

The coherent phase evolution acting on subsystem A is represented through the unitary operator

$$U_\theta = E_\theta \otimes I, \tag{C.10}$$

where

$$E_\theta = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix}. \tag{C.11}$$

The corresponding unitary channel is

$$\mathcal{U}_\theta(\rho) = U_\theta \rho U_\theta^\dagger. \tag{C.12}$$

Local isotropic depolarization acting independently on the two subsystems is described through

$$\mathcal{E}_{p_A,p_B} = \mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}, \tag{C.13}$$

where the single-qubit depolarizing channel satisfies

$$\mathcal{D}_p(I) = I, \tag{C.14}$$

and

$$\mathcal{D}_p(\sigma_i) = (1-p)\sigma_i, \qquad i \in \{x, y, z\}. \tag{C.15}$$

The objective is to prove that

$$\mathcal{E}_{p_A,p_B} \circ \mathcal{U}_\theta = \mathcal{U}_\theta \circ \mathcal{E}_{p_A,p_B}. \tag{C.16}$$

Pauli-Basis Expansion

Consider a generic two-qubit density operator expanded in the Pauli basis:

$$\rho = \frac{1}{4} \sum_{\mu,\nu=0}^{3} S_{\mu\nu}\, \sigma_\mu \otimes \sigma_\nu, \tag{C.17}$$

where

$$\sigma_0 = I,$$

and

$$\sigma_1 = \sigma_x, \qquad \sigma_2 = \sigma_y, \qquad \sigma_3 = \sigma_z.$$

Since both channels are linear, it is sufficient to study their action on generic Pauli-basis elements

$$\sigma_\mu \otimes \sigma_\nu.$$

Action of the Unitary Channel

The local phase unitary acts only on subsystem A. Therefore,

$$\mathcal{U}_\theta \left(\sigma_\mu \otimes \sigma_\nu\right) = \left(E_\theta \sigma_\mu E_\theta^\dagger\right) \otimes \sigma_\nu. \tag{C.18}$$

For $\mu = 0$, the identity operator is preserved trivially:

$$E_\theta I E_\theta^\dagger = I. \tag{C.19}$$

For $\mu \in \{1, 2, 3\}$, the transformed operator admits a Pauli-basis expansion of the form

$$E_\theta \sigma_\mu E_\theta^\dagger = a\sigma_0 + b\sigma_1 + c\sigma_2 + d\sigma_3. \tag{C.20}$$

However, unitary conjugation preserves tracelessness:

$$\operatorname{Tr}\left(E_\theta \sigma_\mu E_\theta^\dagger\right) = \operatorname{Tr}(\sigma_\mu) = 0, \qquad \mu \in \{1, 2, 3\}. \tag{C.21}$$

Since

$$\operatorname{Tr}(\sigma_0) = 2, \qquad \operatorname{Tr}(\sigma_i) = 0 \quad (i = 1, 2, 3),$$

Eq. (C.20) implies

$$a = 0.$$

Therefore,

$$E_\theta \sigma_\mu E_\theta^\dagger = b\sigma_1 + c\sigma_2 + d\sigma_3, \qquad \mu \in \{1, 2, 3\}. \tag{C.22}$$

The unitary evolution therefore preserves the traceless Pauli subspace.

Comparison of the Two Channel Orders

Consider first the ordering

$$\mathcal{E}_{p_A, p_B} \circ \mathcal{U}_\theta. \tag{C.23}$$

Acting on a generic basis element gives

$$
\begin{aligned}
&\mathcal{E}_{p_A,p_B}\left[\mathcal{U}_\theta\left(\sigma_\mu \otimes \sigma_\nu\right)\right] \\
&= \mathcal{D}_{p_A}\left(E_\theta \sigma_\mu E_\theta^\dagger\right) \otimes \mathcal{D}_{p_B}\left(\sigma_\nu\right).
\end{aligned} \tag{C.24}
$$

Using Eq. (C.22), the first factor contains only traceless Pauli operators. Consequently,

$$
\begin{aligned}
&\mathcal{D}_{p_A}\left(E_\theta \sigma_\mu E_\theta^\dagger\right) \\
&= (1-p_A)\left(E_\theta \sigma_\mu E_\theta^\dagger\right), \qquad \mu \in \{1,2,3\}.
\end{aligned} \tag{C.25}
$$

Now consider the opposite ordering:

$$
\mathcal{U}_\theta \circ \mathcal{E}_{p_A,p_B}. \tag{C.26}
$$

Its action gives

$$
\begin{aligned}
&\mathcal{U}_\theta\left[\mathcal{E}_{p_A,p_B}\left(\sigma_\mu \otimes \sigma_\nu\right)\right] \\
&= E_\theta \mathcal{D}_{p_A}(\sigma_\mu) E_\theta^\dagger \otimes \mathcal{D}_{p_B}(\sigma_\nu).
\end{aligned} \tag{C.27}
$$

Using Eq. (C.15),

$$
\begin{aligned}
&E_\theta \mathcal{D}_{p_A}(\sigma_\mu) E_\theta^\dagger \\
&= (1-p_A) E_\theta \sigma_\mu E_\theta^\dagger.
\end{aligned} \tag{C.28}
$$

Comparing Eqs. (C.25) and (C.28) shows that the two orderings coincide.

Therefore,

$$
\mathcal{E}_{p_A,p_B} \circ \mathcal{U}_\theta = \mathcal{U}_\theta \circ \mathcal{E}_{p_A,p_B}. \tag{C.29}
$$

Physical Interpretation

The commutation property derived above follows from two structural features of the effective model:

- the isotropic depolarizing channel acts uniformly on the traceless Pauli subspace;
- unitary conjugation preserves the traceless structure of Pauli operators.

Consequently, coherent phase evolution rotates the polarization-correlation structure without modifying its magnitude, while isotropic local depolarization contracts the correlation amplitudes uniformly without introducing preferred directions in polarization space.

Within the isotropic approximation adopted in the present work, the order of coherent phase evolution and local depolarization is therefore irrelevant.

This property considerably simplifies the analytical structure of the composite phase–depolarization fiber model introduced in Section 5.4, since coherent phase evolution and isotropic local depolarization may be treated independently while remaining mathematically equivalent under composition.

C.3 Statistical Composite Fiber-Channel Analytics

This appendix derives the analytical expressions associated with the statistical composite fiber model introduced in Section 5.5.

The model combines:

- statistical phase averaging,
- local isotropic depolarization acting independently on the two propagation arms.

The reference Bell state is

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}} \left(|01\rangle + |10\rangle\right),$$

with density operator

$$\rho_0 = |\Psi^+\rangle\langle\Psi^+|.$$

The phase-evolved pure state is

$$|\psi(\theta)\rangle = \frac{1}{\sqrt{2}} \left(|01\rangle + e^{i\theta}|10\rangle\right). \tag{C.30}$$

Define the phase-coherence parameter

$$\mu = \left\langle e^{i\theta} \right\rangle = \sum_k p_k e^{i\theta_k}, \tag{C.31}$$

and the depolarization contraction factor

$$\eta = (1 - p_A)(1 - p_B). \tag{C.32}$$

Ensemble density operator

The ensemble-averaged density operator is

$$\rho^{(\text{ens})} = \sum_k p_k |\psi(\theta_k)\rangle\langle\psi(\theta_k)|. \tag{C.33}$$

Using Eq. (C.30), one obtains

$$\rho^{(\text{ens})} = \frac{1}{2}\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & \mu^* & 0 \\ 0 & \mu & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{C.34}$$

Including local isotropic depolarization contracts the bipartite correlations by the factor η. The resulting effective state is

$$\rho_\eta^{(\text{ens})} = \frac{1}{4}\left[I \otimes I + \sum_{i,j} T_{ij}^{(\text{ens})} \sigma_i \otimes \sigma_j \right], \tag{C.35}$$

where the tensor components are derived below.

Correlation tensor

For the ensemble state without local depolarization, the correlation tensor is

$$T^{(\text{ens})} = \begin{pmatrix} \mathrm{Re}(\mu) & -\mathrm{Im}(\mu) & 0 \\ \mathrm{Im}(\mu) & \mathrm{Re}(\mu) & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{C.36}$$

Under local isotropic depolarization,

$$T_\eta^{(\text{ens})} = \eta T^{(\text{ens})}, \tag{C.37}$$

thus

$$T_\eta^{(\text{ens})} = \eta \begin{pmatrix} \mathrm{Re}(\mu) & -\mathrm{Im}(\mu) & 0 \\ \mathrm{Im}(\mu) & \mathrm{Re}(\mu) & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{C.38}$$

The tensor components are therefore

$$T_{xx} = \eta\,\mathrm{Re}(\mu), \tag{C.39}$$
$$T_{xy} = -\eta\,\mathrm{Im}(\mu), \tag{C.40}$$
$$T_{yx} = \eta\,\mathrm{Im}(\mu), \tag{C.41}$$
$$T_{yy} = \eta\,\mathrm{Re}(\mu), \tag{C.42}$$
$$T_{zz} = -\eta. \tag{C.43}$$

All remaining tensor components vanish.

Global purity

The global purity is

$$\gamma\left(\rho_{AB}^{(\text{ens})}\right) = \mathrm{Tr}\left[\left(\rho_{AB}^{(\text{ens})}\right)^2\right]. \tag{C.44}$$

Using the Pauli-basis representation,

$$\rho = \frac{1}{4}\left[I \otimes I + \sum_{i,j} T_{ij}\sigma_i \otimes \sigma_j\right], \tag{C.45}$$

together with Pauli orthogonality,

$$\mathrm{Tr}\left[(\sigma_i \otimes \sigma_j)(\sigma_k \otimes \sigma_l)\right] = 4\delta_{ik}\delta_{jl}, \tag{C.46}$$

one obtains

$$\gamma = \frac{1}{4}\left[1 + \sum_{i,j} T_{ij}^2\right]. \tag{C.47}$$

Substituting Eq. (C.38),

$$\sum_{i,j} T_{ij}^2 = \eta^2\left[2|\mu|^2 + 1\right]. \tag{C.48}$$

Therefore,

$$\boxed{\gamma\left(\rho_{AB}^{(\mathrm{ens})}\right) = \frac{1 + \eta^2(1 + 2|\mu|^2)}{4}} \tag{C.49}$$

For $\eta = 1$, one recovers

$$\gamma\left(\rho_{AB}^{(\mathrm{ens})}\right) = \frac{1 + |\mu|^2}{2}. \tag{C.50}$$

Reduced density operators

The reduced density operators remain maximally mixed:

$$\rho_A = \rho_B = \frac{I_2}{2}. \tag{C.51}$$

Therefore,

$$\boxed{\gamma_A = \gamma_B = \frac{1}{2}}. \tag{C.52}$$

Fidelity with respect to $|\Psi^+\rangle$

The Bell-state fidelity is

$$F_{\Psi^+} = \langle\Psi^+|\rho_\eta^{(\mathrm{ens})}|\Psi^+\rangle. \tag{C.53}$$

Using the ensemble density operator,

$$F_{\Psi^+} = \frac{1}{2}\left[1 + \eta\,\mathrm{Re}(\mu)\right]. \tag{C.54}$$

Therefore,

$$\boxed{F_{\Psi^+} = \frac{1 + \eta\,\mathrm{Re}(\mu)}{2}} \tag{C.55}$$

For $\eta = 1$, this reduces to

$$F_{\Psi^+} = \frac{1 + \mathrm{Re}(\mu)}{2}. \tag{C.56}$$

Concurrence

The ensemble density operator belongs to the class of X-states. For a two-qubit X-state,

$$C = 2\max\left(0, |\rho_{23}| - \sqrt{\rho_{11}\rho_{44}}\right). \tag{C.57}$$

For the present model,

$$\rho_{23} = \frac{\eta\mu}{2}, \tag{C.58}$$

while the depolarizing contribution generates

$$\rho_{11} = \rho_{44} = \frac{1-\eta}{4}. \tag{C.59}$$

Therefore,

$$\begin{aligned} C &= 2\max\left[0, \frac{\eta|\mu|}{2} - \frac{1-\eta}{4}\right] \\ &= \max\left[0, \eta|\mu| - \frac{1-\eta}{2}\right]. \end{aligned} \tag{C.60}$$

Hence,

$$\boxed{C = \max\left[0, \eta|\mu| - \frac{1-\eta}{2}\right]} \tag{C.61}$$

CHSH nonlocality

The maximal CHSH parameter is determined by the Horodecki criterion:

$$S_{\max} = 2\sqrt{\lambda_1 + \lambda_2}, \tag{C.62}$$

where λ_1, λ_2 are the two largest eigenvalues of

$$T^T T.$$

Using Eq. (C.38),

$$T^T T = \eta^2 \begin{pmatrix} |\mu|^2 & 0 & 0 \\ 0 & |\mu|^2 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{C.63}$$

The two largest eigenvalues are therefore

$$\lambda_1 = \eta^2, \qquad \lambda_2 = \eta^2|\mu|^2.$$

Thus,

$$\boxed{S_{\max} = 2\eta\sqrt{1+|\mu|^2}} \tag{C.64}$$

Bell-inequality violation requires

$$S_{\max} > 2, \tag{C.65}$$

which gives

$$\boxed{\eta\sqrt{1+|\mu|^2} > 1} \tag{C.66}$$

Summary

Collecting the analytical expressions:

$$\boxed{\begin{aligned}
T_{xx} &= \eta\,\mathrm{Re}(\mu), \\
T_{xy} &= -\eta\,\mathrm{Im}(\mu), \\
T_{yx} &= \eta\,\mathrm{Im}(\mu), \\
T_{yy} &= \eta\,\mathrm{Re}(\mu), \\
T_{zz} &= -\eta, \\
\gamma\left(\rho_{AB}^{(\mathrm{ens})}\right) &= \frac{1+\eta^2(1+2|\mu|^2)}{4}, \\
\gamma_A = \gamma_B &= \frac{1}{2}, \\
F_{\Psi^+} &= \frac{1+\eta\,\mathrm{Re}(\mu)}{2}, \\
C &= \max\left[0, \eta|\mu| - \frac{1-\eta}{2}\right], \\
S_{\max} &= 2\eta\sqrt{1+|\mu|^2}.
\end{aligned}}$$

These expressions provide the analytical reference for the statistical composite fiber-channel model introduced in Section 5.5.

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